

Are galaxies in compact groups special?

Matthieu Tricottet^{1*}, Gary A. Mamon^{2**}, and Eugenia Díaz-Giménez^{3,4***}

¹ 80, rue d'Alésia, 75014 Paris, France

e-mail: matthieu.tricottet@gmail.com

² Institut d'Astrophysique de Paris (UMR 7095: CNRS & Sorbonne Université), 98 bis boulevard Arago, 75014 Paris, France

³ CONICET. Instituto de Astronomía Teórica y Experimental (IATE), Laprida 854, X5000BGR, Córdoba, Argentina

⁴ Universidad Nacional de Córdoba (UNC). Observatorio Astronómico de Córdoba (OAC), Laprida 854, X5000BGR, Córdoba, Argentina

ABSTRACT

Context. Compact groups combine the highest projected galaxy densities with low velocity dispersions, yet most of them lie inside larger groups. Whether their galaxies are special therefore depends on how they compare with the members of ordinary groups, not only with the field.

Aims. We test whether the galaxies of compact groups of four members (CG₄) differ from those of regular groups of four members (RG₄), from the four brightest members of regular groups (Control_{4B}), and from the projected cores formed by the brightest group galaxy and its three nearest projected companions (Control_{4C}).

Methods. For 62 non-split CG₄ systems we combine Galaxy Zoo 1 morphologies, stellar masses, specific star-formation rates, emission-line diagnostics, spectral indices, and seeing-corrected half-light radii. Each control is compared separately with covariate-adjusted models, galaxy- and group-level matching, and comparisons within shared host groups.

Results. The clearest difference is morphological. At fixed stellar mass, redshift, rank, and group properties, CG₄ galaxies, and especially their satellites, are more often classified elliptical: the odds ratios are 2.1, 1.7, and 2.8 against Control_{4B}, Control_{4C}, and RG₄. Matched quartets show the same ordering: the excess of elliptical satellites is significant against Control_{4B} and RG₄, and has the same sign but is not significant against Control_{4C} ($\Delta = 0.11$, $p = 0.08$). Compact-group members occupy the elliptical-rich centres of their host groups, and no additional excess remains at fixed host-centric radius. At fixed morphology the quenched fractions do not differ, and star formation, colour, nuclear activity, magnitude gaps, and sizes show no comparably robust offsets.

Conclusions. Compact groups of four galaxies host a population shifted towards elliptical morphologies, consistent with transformation accumulated in dense group cores and with preprocessing, without a present-day quenching mechanism specific to compact groups.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Given their high apparent densities and low velocity dispersions (Hickson 1982; Hickson et al. 1992), compact groups appear to be ideal sites for the frequent tidal encounters and mergers that alter galaxy masses, luminosities, sizes, morphologies, and star-formation rates (e.g., Mamon 1992). Mendes de Oliveira & Hickson (1994) provided the classic report of an excess of early-type galaxies in compact groups relative to the field (building on the photometric morphologies of Hickson et al. 1989 and the merger-remnant expectations of Zepf & Whitmore 1993), in line with the broader morphology–density relation (Dressler 1980); multi-wavelength studies of compact-group galaxies likewise report altered infrared, UV, star-formation, and stellar-population properties relative to looser environments (Johnson et al. 2007; Tzanavaris et al. 2010; Bitsakis et al. 2011; Coenda et al. 2012), and their H I de-

ficiency has been organised into an evolutionary sequence (Verdes-Montenegro et al. 2001).

The interpretation of such contrasts is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities: apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2020; Zheng & Shen 2021; Taverna et al. 2023). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints. Recent structural studies find evidence for transformation in compact groups, especially non-isolated systems (Montaguth et al. 2023, 2025, 2026a,b); Montaguth et al. (2026b) compared non-isolated compact-group galaxies with surrounding host populations, but not with ordinary-group quartets reproducing compact-group membership, luminosity-ranking, or projected-core selection.

* ORCID: <https://orcid.org/0009-0008-0062-4450>

** ORCID: <https://orcid.org/0000-0001-8956-5953>

*** ORCID: <https://orcid.org/0000-0001-5374-4810>

In our previous work (Tricottet et al. 2025), we found that, once the direct consequences of the compact-group selection are accounted for, compact groups of four galaxies have group-scale properties broadly similar to those of regular groups of four galaxies and to the eligible cores of regular groups. Here we ask how the galaxy-population contrast changes across three ordinary-group comparison samples: genuine four-member groups (RG₄), the four brightest eligible members of richer groups (Control_{4B}), and BGG-centred projected cores (Control_{4C}), the closest analogue of a compact configuration. We adjust for stellar mass, redshift, galaxy rank, and group-scale properties, and combine individual-galaxy, group-composition, and shared-host contrasts. Section 2 describes the samples, the classifications, and the statistical approach; Sect. 3 presents the results; Sect. 4 discusses their physical interpretation; and Sect. 5 concludes.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study (Tricottet et al. 2025). Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018): reliable HMCs with exactly four members, all present in the Lim–Tempel catalogue, satisfying the volume- and luminosity-complete limits $0.005 \leq z_{\text{BGG}} \leq 0.0452$ and $M_{r,\text{BGG}} \leq -21.81$, with all four galaxies lying within 3 magnitudes of the brightest group galaxy (BGG). In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021), which reduces the compact-group sample from 78 to 62 groups. Following Zheng & Shen (2021), the remaining systems are *isolated* when the four members constitute their whole Lim–Tempel group, *pre-dominant* when they contribute at least half of the host-group *r*-band luminosity, and *embedded* otherwise (6, 37, and 19 systems).

The three control samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least four members and at least three satellites within 3 magnitudes of the BGG, and serve different purposes (Fig. 1). RG₄ contains regular groups with exactly four members and is the most direct equal-membership comparison. Control_{4B} contains the four brightest eligible members of each regular group and tests whether the CG₄ population resembles the luminous core of an ordinary group. Control_{4C} contains the BGG and its three closest projected companions among members within 3 magnitudes of the BGG, reconstructed here from the eligible parent groups by applying the magnitude criterion before ranking companions by projected distance; it is the most compact-core-like analogue, but not a compactness-matched sample: the CG₄–Control_{4C} overlap coefficient in $-\log R_{\text{pair,med}}$ is 0.34, and 18% of CG₄ groups lie above the Control_{4C} 95th percentile in compactness. Every control group containing at least one CG₄ galaxy is excluded, so no galaxy appears on both sides of any comparison; Ap-

Table 1. Final samples. $R_{\text{pair,med}}$ is the median of the six pairwise projected member separations of each quartet (proper kpc, Planck 2015 angular-diameter distances); the table gives the median over groups and the interquartile range.

Sample	Groups	Galaxies	$R_{\text{pair,med}}$ (kpc)
CG ₄	62	248	143 (114–177)
Control _{4B}	698	2792	447 (323–604)
Control _{4C}	703	2812	293 (187–436)
RG ₄	56	224	430 (292–494)

pendix C shows that this exclusion removes the hosts of 51 of the 56 non-isolated compact groups, whose members form the would-be Control_{4C} cores of those hosts. We also remove the systems associated with Lim group 3688, whose single outlying-redshift member ($z = 0.048$, against 0.032–0.034 for the other six) enters both the Control_{4B} and Control_{4C} quartets and inflates their velocity dispersions to $\approx 2000 \text{ km s}^{-1}$. Table 1 summarises the final samples; the Control_{4B} and Control_{4C} group counts differ because the luminosity-ranked and distance-ranked definitions retain different Lim groups after the compact-group exclusion.

We use SDSS DR16 (Ahumada et al. 2020) to retrieve stellar masses, star formation rates (SFRs; Brinchmann et al. 2004; Kauffmann et al. 2003b), emission-line measurements, and Galaxy Zoo morphologies (Lintott et al. 2008, 2011); Appendix A lists the tables and joins. Throughout, ‘SDSS non-AGN reference’ denotes the AGN-filtered reference sample used for calibration and orientation: spectroscopic galaxies with $0.005 \leq z \leq 0.0452$, extinction-corrected Petrosian $r \leq 17.77$, a valid MPA-JHU stellar mass, and finite Galaxy Zoo vote fractions after the AGN filter. It is a field-dominated reference population, not a control sample. All projected distances adopt the Planck 2015 cosmology (Planck Collaboration et al. 2016), as in our previous paper.

2.2. Classifications

Star-formation classes are based on SDSS total specific star-formation rates. The inherited DR12 group catalogues store $\log \text{sSFR} = \text{sfr_tot_p50} - \log m_{\text{tot_p50}}$. In the MPA-JHU catalogues, –9999 is the missing-value sentinel of the `galSpecExtra` table (Brinchmann et al. 2004; Kauffmann et al. 2003b), so rows with unavailable components are treated as *missing data*: they are excluded from the classification and from every sSFR-based figure and fraction, and are reported as counts only (Table 2). All quenched and star-forming fractions are computed among classified galaxies. The missing-sSFR incidence is higher in CG₄ (7% of members, 10% of BGGs) than in the reference sample (1%). Tracing the stored spectrum identifiers shows why: 332 of the 394 missing rows are BOSS-instrument spectra, which the MPA-JHU tables (legacy reductions only) never covered, 60 are legacy spectra whose SFR fit returned the sentinel, and 2 have no spectrum identifier; the missing rows have higher, not lower, median signal-to-noise than the classified ones (Appendix A). Galaxies with a measured sSFR are split into quenched and star-forming classes in the $\log M_{\star}$ – $\log \text{sSFR}$ plane; the boundary and the AGN handling are given in Appendix B (Fig. B.1).

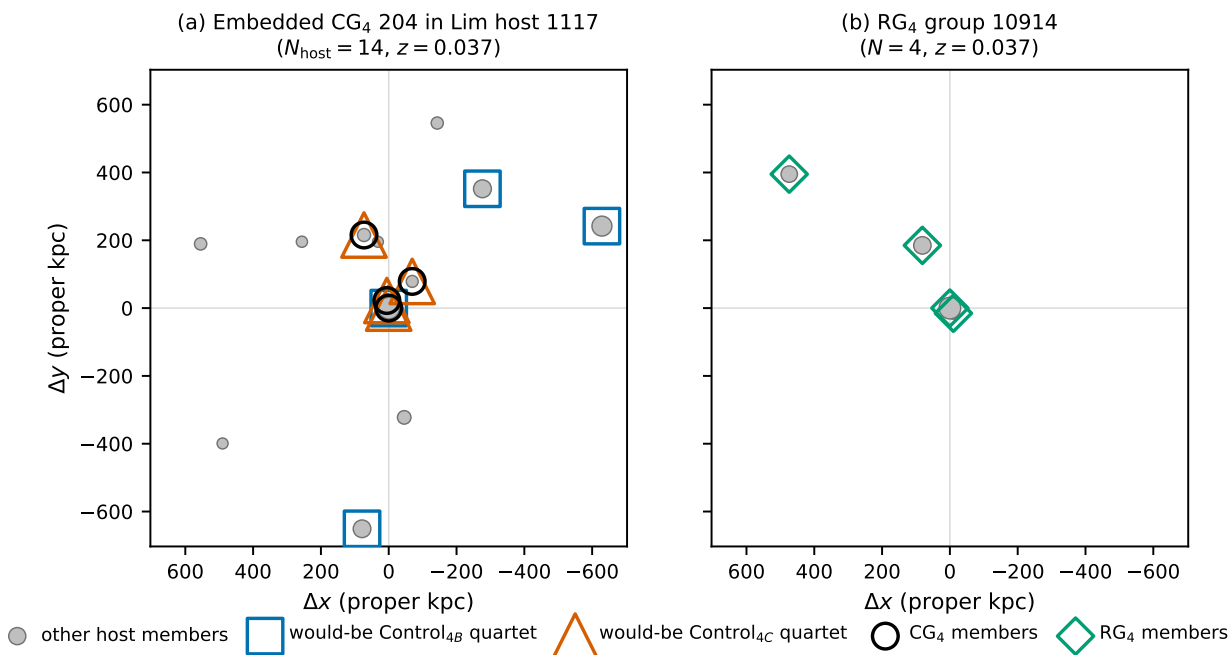


Fig. 1. Sample definitions on the sky. (a) All 14 members of Lim group 1117, which hosts the embedded compact group CG₄ 204 ($z = 0.037$, close to the CG₄ median of 0.037), in projected proper kpc relative to the host BGG; symbol area scales with r -band luminosity. The compact-group members (black circles), the would-be Control_{4B} quartet (the BGG and its three brightest members within 3 mag; blue squares) and the would-be Control_{4C} quartet (the BGG and its three nearest projected members within 3 mag; orange triangles) are marked. Because this host contains CG₄ galaxies it is excluded from the actual control samples. (b) The RG₄ group 10914 ($z = 0.037$) on the same scale.

Morphologies are drawn from Galaxy Zoo 1 (Lintott et al. 2008, 2011), whose vote fractions, de-biased for the redshift-dependent classification bias (Bamford et al. 2009), are provided in zooSpec as `p_el_debiased` (elliptical) and `p_cs_debiased` (combined spiral). We classify a galaxy as elliptical (E) when `p_el_debiased` exceeds 0.5, as spiral (S) when `p_cs_debiased` exceeds 0.5, as uncertain when both vote fractions are finite but neither threshold is exceeded; rows with a missing vote fraction are treated as missing Galaxy Zoo data. Galaxy Zoo 1 does not separate ellipticals from lenticulars, so our E class is a smooth, early-type class rather than a classical-elliptical class. A cross-match with the deep-learning morphologies of Domínguez Sánchez et al. (2018) (DS18; 2436 of 3857 unique galaxies) quantifies this: of the 942 matched E galaxies, 92% have DS18 T Type ≤ 0 , 54% are early with $P_{S0} > 0.5$, and 29% with $P_{S0} \geq 0.8$; conversely, DS18 early types with $P_{S0} > 0.5$ ($N = 730$) fall 70% in our E class and 19% in S (63% and 24% for $P_{S0} \geq 0.8$), with the E share dropping from 77% for face-on to 47% for inclined systems (DS18 $P_{\text{edge-on}} < 0.1$ versus ≥ 0.5), as expected if inclined S0s expose their discs. Throughout, “elliptical” and “E” therefore denote this smooth early-type proxy, which contains a substantial S0 population. We also retrieve the MPA-JHU age-sensitive indices D_n4000 and $H\delta_A$ (Kauffmann et al. 2003b) for the stored spectra of the group galaxies, for descriptive use only (Sect. 3.5, Appendix H.3).

2.3. Galaxy sizes

Our primary size measure is the circular half-light radius of the pure Sérsic fit in the r band, $R_{\text{chl},r}$, from the bulge+disk decomposition catalogue of Simard et al. (2011): PSF-convolved GIM2D fits with re-derived sky levels and deblending designed for crowded fields, so the sizes are corrected for seeing. The catalogue is keyed to SDSS DR7 identifiers; we bridge our DR16 objects to their DR7 counterparts through the SkyServer SpecDR7 cross-match table. Rows whose Simard redshift differs from the catalogue redshift by more than 0.005 are treated as wrong matches and dropped (9 rows). The Simard sizes, tabulated in kpc under the catalogue’s own cosmology, are converted back to angular sizes with the catalogue plate scale and then to proper kpc with the same Planck 2015 conversion as every other projected quantity.

As a secondary, model-independent measure we use the SDSS Petrosian radii `petroR50_r` and `petroR90_r`, which also define the concentration index $C = R_{90,r}/R_{50,r}$ and provide the like-for-like bridge to Coenda et al. (2012). Petrosian radii are not corrected for seeing, and the Petrosian aperture misses roughly 20% of the flux of an $n = 4$ profile, which biases $R_{50,r}$ low for concentrated galaxies (Blanton et al. 2001; Graham et al. 2005). All sizes are required to lie between 0.1 and 50 kpc after re-conversion; Sérsic fits pegged at the GIM2D bounds ($n_g \leq 0.55$ or $n_g \geq 7.9$; 60 galaxies) are excluded from the primary sample, a cut that removes 2.5% of the Simard-matched CG₄ members against 1.0–1.5% in the controls and is therefore conservative with respect to a CG₄ early-type excess; Petrosian entries must satisfy $R_{50,r} > 0$, $R_{90,r} > R_{50,r}$, and a finite positive

uncertainty. The completeness of both measures is given in Appendix H.3.

2.4. Statistical approach

We organise the inference around one question: after accounting for stellar mass, redshift, galaxy rank, and group-scale differences, do compact groups retain a distinct galaxy population? The three controls are pre-specified, physically distinct comparison samples, so CG_4 is compared with each of them separately, and the results are reported as separate contrasts rather than as repeated tests of one null hypothesis. Galaxy-level logistic models give the adjusted population contrast (elliptical versus spiral among usable Galaxy Zoo classifications, quenched versus star-forming among valid sSFRs); group-level matching tests whether whole quartets contain a different satellite mix; and within-host models ask whether compact-subconfiguration members differ from their non-compact co-members. The covariates are stellar mass, redshift, rank, group luminosity, and the quartet velocity dispersion σ_v ; because σ_v from four tracers is noisy, every analysis that uses it is repeated without it (Appendix I).

The independence unit is the *physical* Lim group: standard errors are clustered by physical group and resampling is blocked by physical group or matched pair, because the four galaxies of one group are not independent. We report two-sided p-values with effect sizes and confidence intervals for the planned morphology, star-formation, matching, host, and size analyses, and reserve multiplicity control for the exploratory batteries. The p-value conventions, the covariate-completeness rule, the overlap coefficient, and the treatment of the overlapping control samples in pooled summaries are collected in Appendix D.

3. Results

3.1. Morphology

CG_4 galaxies are more often classified elliptical than the galaxies of every control. Tables 2 and 3 give the raw class counts; among galaxies with a usable E or S classification, the CG_4 elliptical fraction is 59% against 47%, 48%, and 32% in $Control_{4B}$, $Control_{4C}$, and RG_4 (Fisher exact tests: $p = 6.8 \times 10^{-4}$, $p = 1.9 \times 10^{-3}$, and $p < 10^{-4}$). The 38 CG_4 rows with uncertain or missing Galaxy Zoo classifications are excluded rather than assigned to either class. The shift is carried by the satellites: among usable BGG classifications the elliptical fraction is 67.9% in CG_4 and 44.7% in RG_4 (Fisher $p = 1.00$, 1.00, and 0.03 against $Control_{4B}$, $Control_{4C}$, and RG_4), and restricted to star-forming galaxies the E/S mix is compatible with every control (two-sided Fisher exact test: $p = 0.332$, $p = 0.468$, $p = 0.226$), so the raw excess is not carried by star-forming ellipticals.

The excess is present at every well-populated stellar mass (Fig. 2). Between $\log(M_\star/M_\odot) = 9.5$ and 11, the fraction of CG_4 galaxies classified elliptical lies above that of each control, with $Control_{4C}$ the closest comparison, and the mean debiased elliptical votes (open symbols) follow the same ordering; the quenched fractions of satellites and BGGs, by contrast, cross between the samples. The morphology contrast is therefore not a mass effect, and the mass-binned view shows no CG_4 -specific quenching pattern.

Table 2. Number of galaxies in each sSFR class for each sample. Galaxies without a valid sSFR estimate (‘No sSFR’) are excluded from the classification: their percentages are quoted with respect to all galaxies of the (sub)sample, while the quenched and star-forming percentages are quoted with respect to classified galaxies only. The SDSS non-AGN reference extends to substantially fainter absolute magnitudes than the CG_4 sample and is shown for orientation only; it is not a control sample and enters no test in this paper.

Sample	No sSFR	Quenched	Star-forming
CG_4	18 (7%)	146 (63%)	84 (37%)
BGG	6 (10%)	45 (80%)	11 (20%)
Satellites	12 (6%)	101 (58%)	73 (42%)
$Control_{4B}$	180 (6%)	1593 (61%)	1019 (39%)
BGG	58 (8%)	533 (83%)	107 (17%)
Satellites	122 (6%)	1060 (54%)	912 (46%)
$Control_{4C}$	191 (7%)	1560 (60%)	1061 (40%)
BGG	59 (8%)	537 (83%)	107 (17%)
Satellites	132 (6%)	1023 (52%)	954 (48%)
RG_4	5 (2%)	101 (46%)	118 (54%)
BGG	3 (5%)	38 (72%)	15 (28%)
Satellites	2 (1%)	63 (38%)	103 (62%)
SDSS ref.	634 (1%)	7705 (15%)	44192 (85%)

Table 3. Number of galaxies in each Galaxy Zoo vote-fraction morphology class for each sample. Percentages are quoted with respect to each sample’s final row count; the inferential E/S comparison conditions on the usable E/S subset. The No GZ column counts rows without finite Galaxy Zoo vote fractions after merging and is separated from the finite-vote uncertain class. As in Table 2, the SDSS non-AGN reference is not a control sample.

Sample	Ellip.	Spiral	Unc.	No GZ
CG_4	124 (50%)	86 (35%)	7 (3%)	31 (12%)
$Control_{4B}$	1106 (40%)	1266 (45%)	116 (4%)	304 (11%)
$Control_{4C}$	1118 (40%)	1229 (44%)	144 (5%)	321 (11%)
RG_4	60 (27%)	129 (58%)	15 (7%)	20 (9%)
SDSS ref.	10091 (19%)	34715 (66%)	7725 (15%)	0 (0%)

The excess survives adjustment for galaxy and group properties. In logistic models of elliptical-versus-spiral classification with stellar mass, redshift, rank, group luminosity, and σ_v as covariates (standard errors clustered by physical group), the CG_4 odds ratio is 2.14 (95% CI [1.58, 2.90]) against $Control_{4B}$, 1.67 (95% CI [1.25, 2.24]) against $Control_{4C}$, and 2.81 (95% CI [1.79, 4.38]) against RG_4 (Wald tests: $p < 10^{-4}$, $p = 5.9 \times 10^{-4}$, $p < 10^{-4}$; Fig. 3). The ordering is physically suggestive: the excess is largest against the two ~ 440 kpc environments ($Control_{4B}$, RG_4) and smallest against the ~ 293 kpc projected cores ($Control_{4C}$), the control that most resembles a compact configuration, although the three estimates are not tested against one another; refitting without the noisy σ_v covariate moves the odds ratios by less than their standard errors (Appendix I).

The signal is carried by satellites. In the pooled secondary model over the deduplicated control catalogues (Appendix H), the satellite elliptical odds ratio is 2.08 (95% CI [1.51, 2.88]) ($p < 10^{-4}$), whereas the BGG estimate is 0.78 (95% CI [0.39, 1.53]) ($p = 0.464$); the pooled all-galaxy odds ratio is 1.77 ($p = 1.4 \times 10^{-4}$).

Whole compact groups contain more elliptical satellites than matched ordinary groups. Treating each quartet as the unit, we count its elliptical satellites ($N_{E,sat} \in \{0, 1, 2, 3\}$ among classified satellites) and match CG_4 groups one-to-one to controls on redshift, BGG stellar mass, total quartet luminosity, and σ_v (propensity caliper 0.2 SD of the logit),

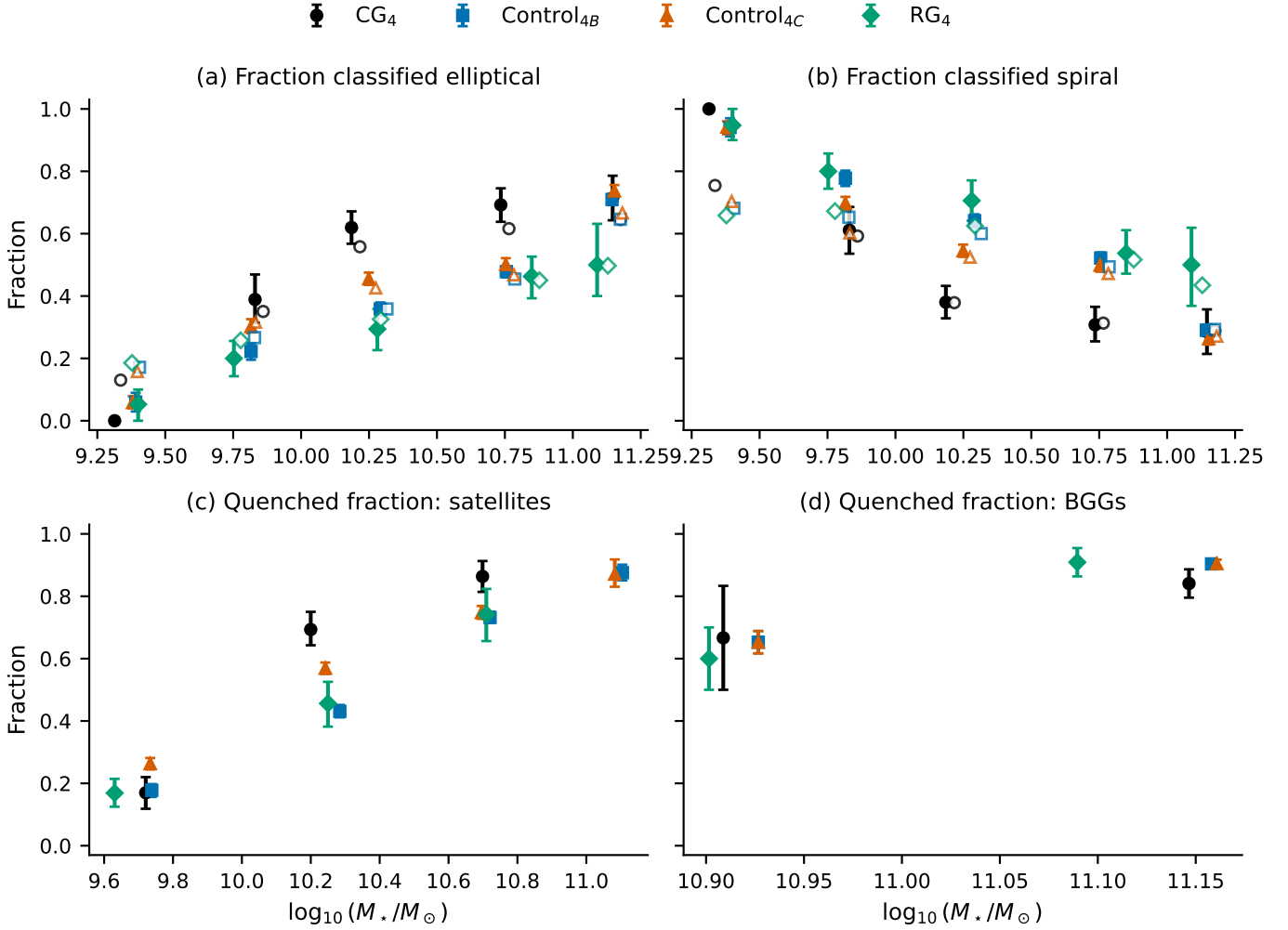


Fig. 2. Morphology and quenched fractions versus stellar mass. Top: fraction of galaxies classified elliptical (a) and spiral (b) among usable Galaxy Zoo classifications (filled symbols) and mean debiased elliptical and spiral vote fractions over all galaxies with finite votes (open symbols, offset slightly in mass). Bottom: quenched fractions among valid-sSFR satellites (c) and BGGs (d). All samples use the same bins within each row: edges 7.0, 9.5, 10.0, 10.5, 11.0, 12.5 for morphology and 7.0, 10.0, 10.5, 11.0, 12.5 for quenching. Each estimate is plotted at the median stellar mass of the contributing galaxies; points are not connected because the bins have unequal widths. Bars are 16th–84th percentile physical-group-blocked bootstrap intervals of the filled symbols; bins with fewer than five galaxies or three groups are omitted.

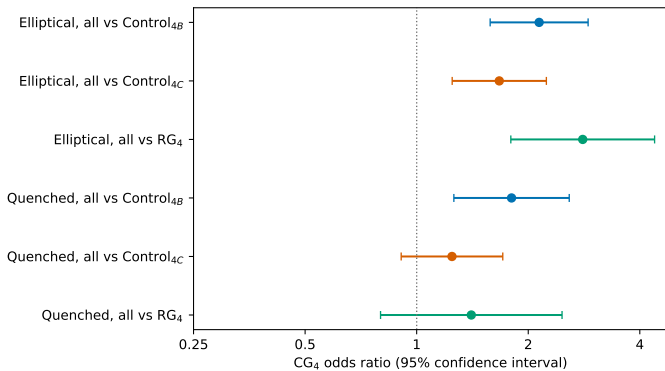


Fig. 3. Adjusted CG₄ odds ratios from the three separate per-control contrasts (colour-coded as in Fig. 2) for the all-galaxy elliptical and quenched outcomes. Bars show 95% confidence intervals; the vertical line marks equal odds.

separately for each control (Table 4). The CG₄-minus-control difference in elliptical-satellite fraction is $\Delta = 0.16$ against Control_{4B} (95% CI [0.03, 0.28]; paired sign-flip permutation $p = 0.016$) and 0.20 against RG₄ ($p = 0.026$), while against the projected cores Control_{4C} it has the same sign but is not significant ($\Delta = 0.11$, $[-0.02, 0.24]$, $p = 0.08$). Adding the mean satellite stellar mass to the match preserves the hierarchy ($\Delta = 0.20$, 0.09, and 0.26). Without σ_v in the match the Control_{4B} difference grows to 0.28 because a different set of control quartets, from dynamically cooler hosts, is selected (only 7 of 54 control groups are shared), while the Control_{4C} and RG₄ values change little (Appendix I).

Individually matched galaxies confirm the shift. Matching 234 of the 248 CG₄ galaxies one-to-one to ordinary-group galaxies of the deduplicated control pool on stellar mass, redshift, rank, group luminosity, and σ_v (exact rank strata, common support, 0.2-SD caliper; maximum absolute standardized mean difference 0.08 after matching, Appendix H), the matched elliptical-fraction difference is 0.14

Table 4. Per-control group-level matched satellite-composition contrasts. Δ is the matched mean difference in elliptical-satellite fraction (CG₄ minus control) with its 95% pair-bootstrap interval ($B = 9999$); p_{perm} is the paired sign-flip permutation p-value and p_{Holm} its Holm adjustment across the three controls. Pair counts: 54 CG₄ groups have complete matching covariates and classified satellites; against Control_{4B} and Control_{4C} (630 and 632 available control groups) every CG₄ group finds a within-caliper partner, whereas the RG₄ pool retains 28 pairs on common support. The pooled row (deduplicated controls) is a secondary summary.

Control	Pairs	Δ	95% CI	p_{perm}	p_{Holm}
Control _{4B}	54	0.160	[0.034, 0.284]	0.02	0.05
Control _{4C}	54	0.114	[-0.015, 0.238]	0.08	0.08
RG ₄	28	0.196	[0.036, 0.351]	0.03	0.05
Pooled	54	0.167	[0.040, 0.290]	0.02	—

(95% block-bootstrap interval [0.04, 0.24]; $p = 0.007$), rising to 0.19 for the 130 satellite pairs alone ($p = 0.004$).

Within their own host groups, compact-group members occupy the elliptical-rich centres rather than forming a distinct population. For the 19 embedded and 37 predominant systems, whose quartets live inside richer Lim groups (54 distinct hosts, since None Lim groups each host two compact groups; 958 member galaxies, 224 of them in a CG₄), we compare the CG₄ members with the non-CG₄ members of the same host by conditional logistic regression stratified by host, which removes every host-level property by design, with stellar mass, host luminosity rank, and projected host-centric radius as covariates. The elliptical odds ratio for CG₄ membership is 1.47 (95% CI [0.92, 2.35]; $p = 0.108$; 791 galaxies in 53 informative strata), but rises to 1.90 ([1.23, 2.95]; $p = 0.004$) when the radius covariate is dropped. The difference between the two fits is the pattern expected if CG₄ galaxies preferentially occupy the early-type-rich central regions of their hosts; the radius-adjusted interval still permits moderate effects. A fixed-effects GLM gives the same direction (Appendix I).

3.2. Star formation at fixed morphology

The star-formation contrast is weaker, control-dependent, and largely a consequence of the morphology mix. In the raw comparison, Fisher's exact test on the star-forming fraction gives $p = 0.481$, $p = 0.262$, and $p = 3 \times 10^{-4}$ against Control_{4B}, Control_{4C}, and RG₄ for all galaxies, and $p = 0.302$, $p = 0.114$, and $p = 2.3 \times 10^{-4}$ for satellites (two-sided Fisher exact test), the deficit being clearest against RG₄; the raw RG₄ contrast survives every extreme assignment of the missing-sSFR galaxies (Appendix B). After adjustment for the same covariates as in Sect. 3.1, the quenched odds ratio is 1.80 (95% CI [1.26, 2.58]) against Control_{4B} ($p = 0.001$) but only 1.25 (95% CI [0.91, 1.71]) against Control_{4C} ($p = 0.173$) and 1.40 (95% CI [0.80, 2.47]) against RG₄ ($p = 0.238$; Fig. 3).

At fixed morphology the samples are compatible. The conditional quenched fractions among sSFR-classified galaxies are $P(Q|E) = 0.908, 0.903, 0.887$, and 0.850 for CG₄, Control_{4B}, Control_{4C}, and RG₄ (homogeneity χ^2 test, $p = 0.39$; Table B.2), and a Kitagawa decomposition of each raw CG₄-minus-control quenched-fraction difference into a morphology-mix term and a conditional term leaves the conditional terms consistent with zero: -0.017

$[-0.073, +0.038]$ versus Control_{4B}, $+0.002 [-0.054, +0.058]$ versus Control_{4C}, and $+0.050 [-0.035, +0.136]$ versus RG₄ (group-blocked 95% intervals). The individually matched quenched-fraction difference is likewise small, 0.03 (95% CI $[-0.06, 0.12]$; $p = 0.526$). The quenched odds-ratio pattern is thus what the different morphology mixes imply, with no additional quenching at fixed vote-based morphology. Nor does the interpretation depend on the GMM boundary: with a fixed threshold, $\log_{10}(\text{sSFR}/\text{yr}^{-1}) = -11.0$, the CG₄ and RG₄ satellite star-forming fractions are 41.4% and 60.2% (difference -18.8 points; $p = 5.3 \times 10^{-4}$), so the quenched/star-forming contrast remains the more model-dependent of the two.

Star-forming CG₄ galaxies do not sit off the star-forming main sequence (SFMS). Relative to an order-2 polynomial SFMS fitted to the star-forming galaxies of the SDSS reference sample (Appendix E), the median residual of CG₄ star-forming galaxies exceeds that of the controls by 0.056, 0.058, and 0.046 dex for Control_{4B}, Control_{4C}, and RG₄ (68% group-bootstrap intervals [0.03, 0.10], [0.04, 0.10], $[-0.04, 0.10]$; sign-crossing $p = 0.06$, $p = 0.03$, $p = 0.59$; Fig. 4a), whereas the mean paired difference over the 39 individually matched pairs in which both members are star-forming is -0.15 dex ($[-0.31, 0.01]$; $p = 0.06$). The two signs are not in conflict: they are different statistics of different subsets (on the same 39 pairs the median paired difference is -0.08 dex, Wilcoxon $p = 0.26$), and both reflect the mass dependence of the residuals about the field-fitted relation: control star-forming galaxies above $\log(M_{\star}/M_{\odot}) = 10.3$ lie $\simeq 0.2$ dex below it while the 23 massive star-forming CG₄ galaxies do not, which raises the raw medians, whereas the matched star-forming controls are less massive than their CG₄ partners. We therefore treat SFMS residuals as a secondary diagnostic. Quenched CG₄ galaxies lie slightly below the quenched sequence of the reference sample (Fig. 4b; order-4 fit, median offsets -0.07 , -0.06 , and -0.15 dex against Control_{4B}, Control_{4C}, and RG₄; $p = 0.002$, $p = 0.013$, $p = 2 \times 10^{-4}$); MPA-JHU sSFRs of quenched galaxies are largely D_n4000 -calibrated, so these residuals are weakly informative.

3.3. Compact-group classes

The elliptical excess belongs to the compact groups that live inside richer hosts. Figure 5 shows the satellite elliptical fraction $f_E = N_E/(N_E + N_S)$ for the three Zheng-Shen classes of Sect. 2.1: $f_E = 0.638^{+0.122}_{-0.143}$ for the 19 embedded and $0.562^{+0.095}_{-0.100}$ for the 37 predominant systems, against $0.286^{+0.261}_{-0.169}$ for the 6 isolated ones (Wilson 95% intervals; Table F.2). The isolated compact groups are compatible with the loose RG₄ groups (group-level permutation test on the per-group mean satellite f_E , $p = 1.00$) and sit below the rest of CG₄ ($p = 0.023$), whereas the two embedded-type classes are indistinguishable from each other (predominant-versus-embedded odds ratio at fixed stellar mass 0.97 , $p = 0.94$). The host halo masses differ strongly between the classes (median $\log M_{200c} = 12.86, 13.83$, and 13.13 for isolated, embedded, and predominant systems), yet adding $\log M_{200c}$ to the class model changes neither the class coefficients nor the conclusion ($\log M_{200c}$ odds ratio 0.80 per standard deviation, $p = 0.31$); with only 6 isolated groups the class contrast itself is exploratory, and an exploratory battery of other group-scale descriptors is

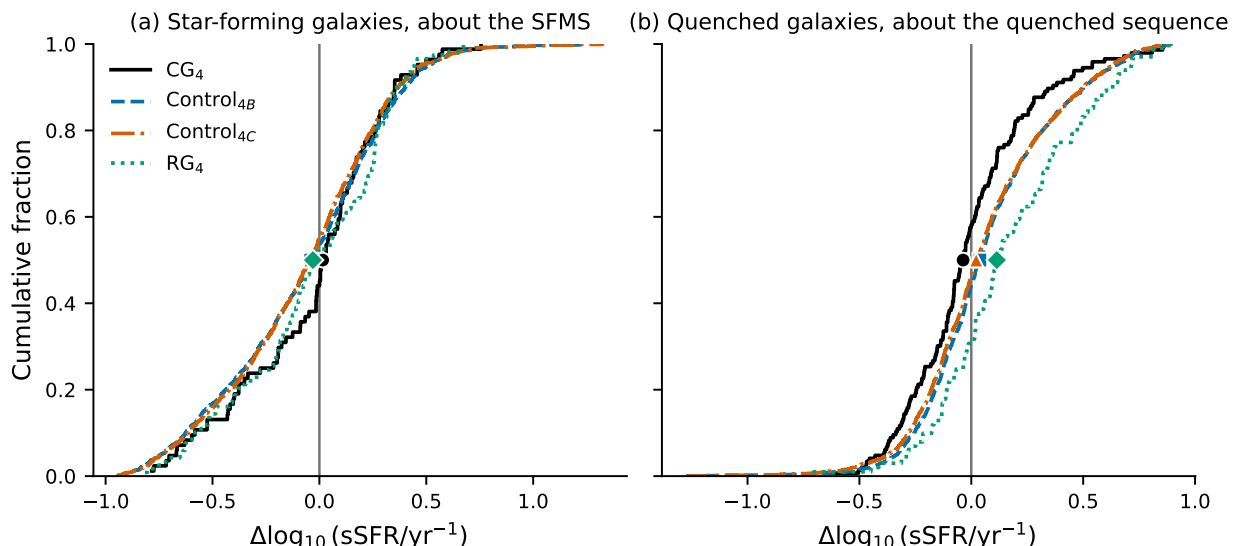


Fig. 4. Cumulative distributions of the sSFR residuals of (a) star-forming galaxies about the order-2 star-forming main sequence and (b) quenched galaxies about the order-4 quenched sequence, both fitted to the SDSS non-AGN reference sample. Zero marks the fitted relation; filled symbols mark the sample medians. MPA-JHU sSFRs of quenched galaxies are largely D_n4000 -calibrated, so the residuals in panel (b) are weakly informative.

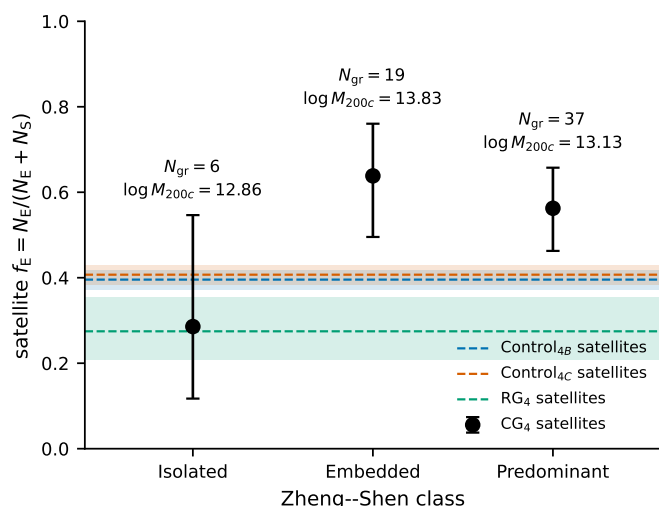


Fig. 5. Satellite elliptical fraction of the CG_4 groups by Zheng–Shen class (points, Wilson 95% intervals), annotated with the number of groups and the median host $\log M_{200c}$ of each class. The dashed lines and bands are the satellite elliptical fractions of the three control samples with their Wilson intervals.

given in Appendix G. In Tricottet et al. (2025) the embedded and predominant labels were interchanged relative to the definition of Zheng & Shen (2021) adopted here, which affects the class counts and host masses quoted there but none of the isolated-group results.

3.4. Local projected density

Local projected density accounts for part, but not a well-determined part, of the excess. For each of the 6076 catalogue rows we compute a pairwise tidal index $T_i = \sum_j M_{\star,j} / R_{ij}^3$ over the selected quartet co-members (R_{ij} in proper kpc; co-members without a stellar mass contribute

zero, so T_i is a lower bound in those groups) and refit the elliptical model, adjusted for stellar mass and satellite status and clustered by physical group, with and without the index on the three control catalogues pooled *without* objid deduplication (unlike the pooled models of Appendix H). The CG_4 odds ratio falls from 1.95 ($p < 10^{-4}$) to 1.39 ($p = 0.031$) when the index is added, the index itself having an odds ratio of 1.42 per 1.07-dex standard deviation of $\log T_i$ ($p < 10^{-4}$). Because the index is built from the same projected separations that define a compact group, conditioning on it partly conditions on the selection, so this attenuation is not evidence that local density mediates the morphology contrast. Nor is the residual CG_4 term robust on common support: the CG_4 and control $\log T_i$ distributions overlap with coefficient 0.54 but their medians differ by 1.1 dex, and the residual odds ratio becomes 1.40 with a spline in $\log T_i$, 1.27 after deduplicating the control pool ($p = 0.12$), and 1.22 on the common-support window ($p = 0.20$). The index is a proxy for immediate neighbours in projection, not a measurement of tidal history; the restriction to quartet co-members is innocuous (Appendix H.3).

3.5. Other galaxy properties

None of the other galaxy properties shows an offset as robust as the morphology excess (Fig. 6). Strong $H\alpha$ emission ($H\alpha EW \leq -3 \text{ \AA}$, emission negative) is rarer in CG_4 : 32% of members against 41%, 43%, and 57% in $Control_{4B}$, $Control_{4C}$, and RG_4 (Fisher $p = 0.01$, $p = 0.002$, $p < 10^{-4}$), but this morphology-blind fraction largely restates the morphology contrast, since LINER-like emission is common in early-type galaxies, and it is not adjusted for the samples' different morphology mixes. The AGN-like fraction among BPT-classified galaxies (above the Kewley et al. 2001 line, or flagged as AGN without line ratios; composites between the Kauffmann et al. 2003a and Kewley et al. 2001 lines) is 38% in CG_4 , and its mass- and environment-adjusted CG_4 odds ratio is 1.17 ($p = 0.36$). The D_n4000 index follows stel-

lar mass in every sample; the CG₄ medians sit above the controls only in the 10.0–10.5 bin ($D_n4000 = 1.77$ against 1.59, 1.68, and 1.56), where the CG₄ elliptical fraction also peaks relative to the controls, and coincide with them elsewhere; a post-starburst classification from D_n4000 and $H\delta_A$ is deferred to a follow-up paper.

CG₄ galaxies are not smaller than ordinary-group galaxies of the same stellar mass. With the seeing-corrected Simard radii, the satellite size offset at fixed stellar mass, redshift, and group-scale covariates is -0.010 dex (-2% ; 95% CI $[-0.039, 0.019]$; $p = 0.51$; 2575 satellites in 754 physical groups), and -0.006 dex for all galaxies ($p = 0.65$); the per-control estimates are -0.035 dex against Control_{4B}, 0.000 dex against Control_{4C}, -0.006 dex against RG₄ (Fig. J.1; Appendix J). The one-to-one matched difference is -0.019 dex for the 166 pairs with Simard sizes (95% CI $[-0.064, 0.026]$; $p = 0.40$), and -0.033 dex with the uncorrected Petrosian radii ($p = 0.03$). The Simard measure resolves 77% of the CG₄ galaxies against 71% of the pooled controls, so the Simard coefficients carry a differential-completeness caveat that the Petrosian rerun does not share (Petrosian satellite coefficient -0.019 dex, $p = 0.11$). Colours, magnitude gaps, and projected phase-space trends are likewise control-dependent or weak; they are reported in Appendix F.

3.6. Robustness

The morphology excess does not depend on the vote threshold, on the classifier, or on the crowded closest pairs (Table 5). Excluding every galaxy whose nearest catalogued neighbour lies within 55 arcsec, the SDSS fibre-collision scale, removes 19% of CG₄ members against 6–10% of the controls and moves the adjusted elliptical odds ratios from 2.14 to 2.16 (Control_{4B}), 1.67 to 1.71 (Control_{4C}), and 2.81 to 2.71 (RG₄). Across classification thresholds of 0.4–0.8 the pooled adjusted odds ratio stays between 1.73 and 2.19, the continuous debiased vote fraction p_E itself is offset by 0.089 ($p < 10^{-4}$), and the Galaxy-Zoo-independent Sérsic index ($n_g > 2.5$; 75% coverage) gives an odds ratio of 1.60 ($p = 0.002$). With the DS18 T Type ≤ 0 classification as the outcome, the per-control CG₄ odds ratios are 2.11, 1.49, and 2.22 (all $p \leq 0.017$; $n = 1822, 1913, 324$). The crowding, threshold, and σ_v reruns are detailed in Appendix I.

4. Interpretation and comparison with previous work

4.1. An elliptical satellite excess linked to dense group cores

Compact groups of four galaxies host more elliptical satellites than any of the three ordinary-group comparison samples, and the size of the excess follows the compactness of the comparison: it is largest against the ~ 440 kpc Control_{4B} and RG₄ quartets and smallest against the ~ 293 kpc projected cores Control_{4C} (Sect. 3.1). Within their own hosts, compact-group members show the excess only when their central location is not held fixed, which is the behaviour expected if they occupy the early-type-rich centres of their groups. The within-host estimator refines the surrounding-host comparison of Montaguth et al. (2026b) by asking whether morphology differs beyond shared host and central location; samples, morphology definitions, and structural measurements differ, so this refines rather than contradicts

Table 5. Robustness of the CG₄ morphology contrast. Odds ratios (OR) are adjusted CG₄ coefficients; the continuous row is the adjusted offset in the debiased elliptical vote fraction p_E . P-values are unadjusted.

Test	Result
55-arcsec neighbour exclusion	all-row E: 52.5% / 40.1% (CG ₄ / Control _{4C}); E/S Fisher $p = 0.002$
Add close-neighbour flag to model	CG ₄ OR 1.76 [1.30, 2.39]; $p = 2.6 \times 10^{-4}$
DS18 T Type ≤ 0	OR 2.11 / 1.49 / 2.22 (Control _{4B} / Control _{4C} / RG ₄); all $p \leq 0.017$
Threshold sweep 0.4–0.8	OR 1.73–2.19
Continuous p_E offset	0.089 [0.047, 0.131]; $p < 10^{-4}$
Sérsic $n_g > 2.5$	OR 1.60; $p = 0.002$
Without σ_v	OR 2.22 / 1.68 / 2.76 (Control _{4B} / Control _{4C} / RG ₄)

their result. The scale of the difference depends on what is held fixed: restricting the individually matched comparison to satellites raises the elliptical-fraction difference to 0.19, and present-day stellar mass may itself encode part of a galaxy’s environmental history, so the fixed-mass estimates describe the remaining association rather than the total accumulated transformation.

The local projected-density experiment of Sect. 3.4 is consistent with this picture without proving it: the tidal-index term is itself strongly predictive of elliptical morphology and attenuates the CG₄ coefficient (from 1.95 to 1.39), as expected if compact groups select galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label, are linked to the elliptical population. Because the index is entangled with the selection and its residual term is not stable on common support, the causal ordering of selection, density, and morphology is not identified by projected snapshots. This reading is consistent with compact groups tracing the high-density, high-interaction tail of group environments—some genuinely compact, others dense cores or substructures inside larger haloes—and with the stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work (Montaguth et al. 2023, 2025, 2026a,b). The 6 isolated compact groups are compatible with loose RG₄ groups in satellite elliptical fraction and lie below the rest of CG₄ (Sect. 3.3); this is consistent with a larger-scale host contribution, but the isolated sample is too small to separate the explanations, and the host halo mass adds nothing to the class contrast or to the adjusted models (Appendix H.3).

Stellar-population diagnostics could in principle separate an accumulated enrichment history from a shorter star-formation timescale, but we defer a stellar-metallicity analysis for two reasons. No quenching excess is detected at fixed morphology (Sect. 3.2), so a strangulation-type metallicity test of the kind proposed by Peng et al. (2015) would have little leverage here, and the single-fibre SDSS spectra sample only the central ~ 2 kpc of the galaxies at the sample redshifts, where the gradients of both metallicity and age are steepest.

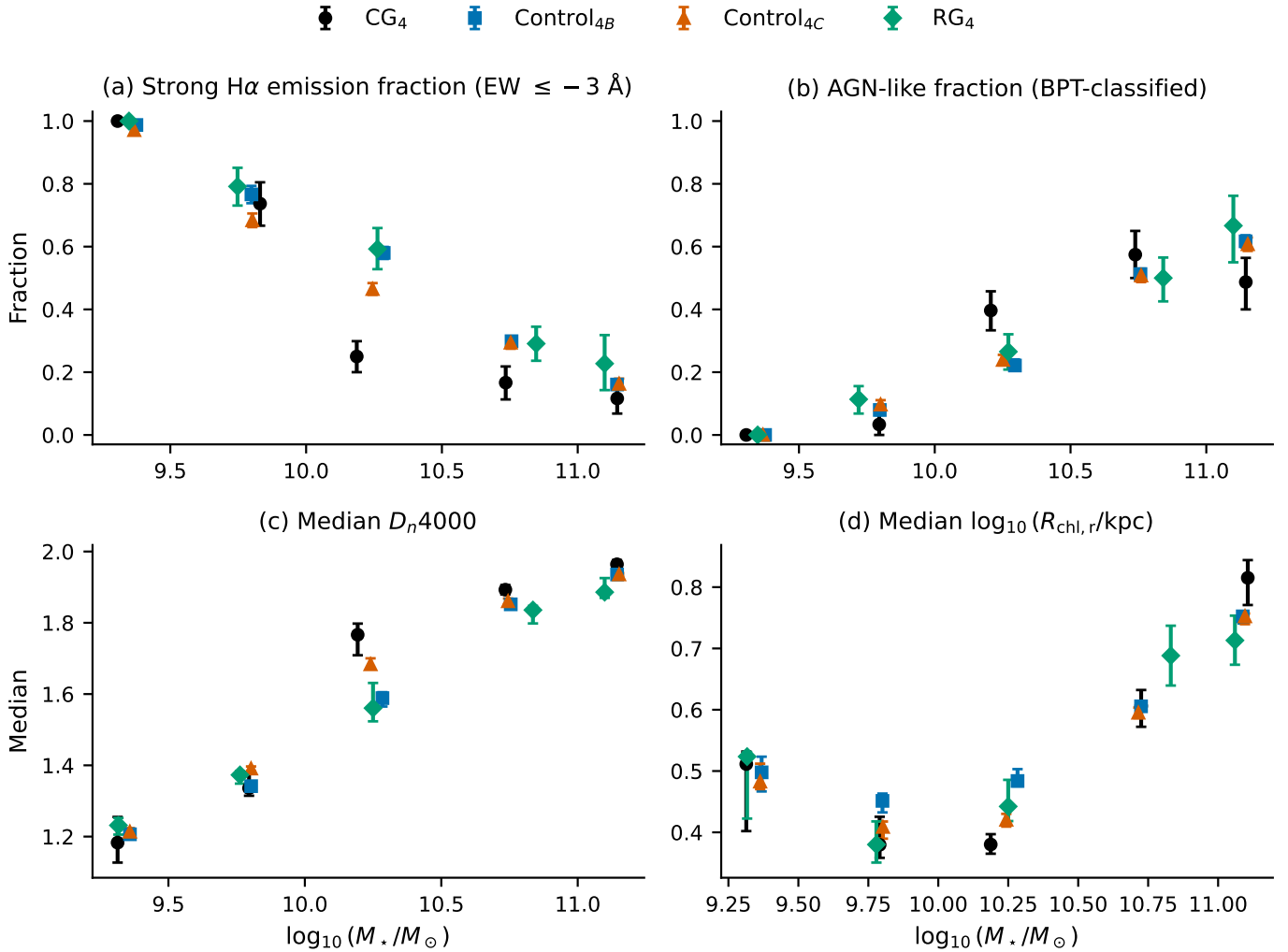


Fig. 6. Other galaxy properties versus stellar mass, for all galaxies of the four samples, with the bins, symbols, and group-blocked 16th–84th percentile intervals of Fig. 2: (a) fraction with strong H α emission ($\text{EW} \leq -3 \text{ \AA}$); (b) AGN-like fraction among BPT-classified galaxies; (c) median D_n4000 ; (d) median seeing-corrected half-light radius $\log_{10}(R_{\text{chl},r}/\text{kpc})$. Descriptive only.

4.2. Elliptical-vote morphologies and transition galaxies

Our E class is a smooth early-type proxy with a substantial S0 content (Sect. 2.2), so the excess is one of early-type-like galaxies rather than of classical ellipticals. Whether the excess is made of S0s, of ellipticals, or of both is a question that Galaxy Zoo 1 cannot answer and that we reserve for a dedicated, pre-registered analysis of the DS18 classes by environment. This caveat matters when comparing with the S-PLUS studies of compact-group transition galaxies and R_e –Sérsic-index trends (Montaguth et al. 2023, 2025, 2026a,b): their structural measurements and transition populations probe pathways, whereas our control hierarchy tests which apparent CG₄ signatures survive across genuine four-member, luminosity-selected, and BGG-centred projected-core quartets.

4.3. Why the star-formation and size signals are weaker

The star-formation contrast is what the morphology mix implies. At fixed vote-based morphology the quenched fractions of the four samples agree and the Kitagawa conditional terms are consistent with zero (Sect. 3.2), so the

quenched odds ratios of Fig. 3 measure the different proportions of ellipticals, not an additional quenching channel; the raw satellite deficit of star formation against RG₄ survives every assignment of the missing-sSFR galaxies but is control-dependent once mass and rank are held fixed. The non-detections are also limited by the spectroscopy. CG₄ has more catalogued galaxies with a nearest neighbour below the 55-arcsec fibre-collision scale than the pooled controls (19% versus 8%), and Control_{4C}, selected as a compact-core analogue, has the highest close-pair fraction of the controls (10%). Additional close companions may therefore be preferentially missing in CG₄, hiding short-lived starbursts or post-starburst phases and biasing the pairwise tidal index downward, and the MPA-JHU aperture corrections (Brinchmann et al. 2004; Salim et al. 2007) may be less secure for interacting or blended members. We therefore read the sSFR, main-sequence, and H α results as conservative non-detections rather than as proof of no star-formation effect.

The size result carries a parallel message. Coenda et al. (2012) found compact-group galaxies to be smaller at fixed luminosity than loose-group and field samples, especially

early types, using SDSS Petrosian half-light radii; against three regular-group controls, with seeing-corrected model sizes, cluster-robust adjusted models, one-to-one matching, and crowding tests, we find no size offset at fixed stellar mass (satellite coefficient -0.010 dex, bounding any mean CG_4 size deficit to roughly 0.04 dex; matched Simard difference -0.019 dex). This is compatible with a compact-group versus field offset if ordinary-group cores have already undergone comparable structural processing. A negative fixed-mass size offset could reflect tidal truncation but also higher bulge dominance or residual blending in crowded fields; the morphology-stratified fits, the Petrosian cross-check, and the crowding exclusions of Appendix J bound both readings.

5. Conclusion

We have tested whether galaxies in CG_4 compact groups remain special after removing split systems, comparing them with RG_4 , $Control_{4B}$, and $Control_{4C}$ through individual-galaxy, group-composition, and shared-host contrasts. The inference applies to this luminous, volume-limited, exactly four-member non-split HMCg sample rather than to all compact groups.

The main result is an elliptical satellite excess. CG_4 galaxies are more often classified elliptical than all three regular-group comparisons after accounting for galaxy and group properties; the excess is strongest against $Control_{4B}$ and RG_4 and weakest against the projected cores $Control_{4C}$, matched quartets and satellite-mass-balanced variants show the same hierarchy, BGGs show no corresponding difference, and within their hosts CG_4 members show no excess beyond their central locations. Vote-threshold, continuous-vote, Sérsic-index, close-pair, and independent DS18 checks support the result, and a projected tidal-density proxy attenuates it in the way expected if local density contributes, although projected data identify neither mediation nor a unique mechanism.

The star-formation result is the morphology mix seen through a different lens: at fixed vote-based morphology the quenched fractions agree, the matched quenched-fraction difference is small, and star-forming CG_4 galaxies show no robust main-sequence offset. At fixed stellar mass the seeing-corrected satellite size estimate is -0.010 dex with an interval including zero, so we do not claim a compact-group size effect. Colours, nuclear activity, magnitude gaps, $H\alpha$ emission, D_n4000 , and projected phase-space trends identify no simple alternative driver.

We therefore interpret CG_4 systems as hosting a population shifted towards smooth, early-type-like morphologies, consistent with structural transformation accumulated in dense group cores and with preprocessing, and without evidence for a compact-group-specific quenching mechanism at the present epoch; the isolated subset is too small to isolate a distinct mode. Explicit E–S0–spiral and transition-galaxy classifications, bulge–disc structure, homogeneous halo masses, and stellar-population diagnostics will be needed to determine whether the satellite excess reflects disc fading, tidal heating, or merger-driven transformation.

Data availability

The input data are public SDSS, MPA-JHU, Galaxy Zoo, Simard et al. (2011), group, and compact-group catalogues cited above. Derived galaxy and group tables will be made available through CDS at publication. The analysis and paper-generation code is available at https://github.com/MatthieuTricottet/CG_Gals; the full SDSS query used for the reference sample is in `src/data_loader.py`, the DR12 spectrum-provenance and spectral-index retrievals in `src/ssfr_quality_audit.py` and `src/spectral_indices.py`, and the read-only diagnostics of this revision under `results/diagnostics/`. The article release, including online diagnostic supplements, will be archived with a DOI at acceptance.

Acknowledgements. Funding for the Sloan Digital Sky Survey IV has been provided by the Alfred P. Sloan Foundation, the U.S. Department of Energy Office of Science, and the Participating Institutions. SDSS acknowledges support and resources from the Center for High-Performance Computing at the University of Utah. The SDSS web site is www.sdss4.org. SDSS is managed by the Astrophysical Research Consortium for the Participating Institutions of the SDSS Collaboration including the Brazilian Participation Group, the Carnegie Institution for Science, Carnegie Mellon University, Center for Astrophysics | Harvard & Smithsonian (CfA), the Chilean Participation Group, the French Participation Group, Instituto de Astrofísica de Canarias, The Johns Hopkins University, Kavli Institute for the Physics and Mathematics of the Universe (IPMU) / University of Tokyo, the Korean Participation Group, Lawrence Berkeley National Laboratory, Leibniz Institut für Astrophysik Potsdam (AIP), Max-Planck-Institut für Astronomie (MPIA Heidelberg), Max-Planck-Institut für Astrophysik (MPA Garching), Max-Planck-Institut für Extraterrestrische Physik (MPE), National Astronomical Observatories of China, New Mexico State University, New York University, University of Notre Dame, Observatório Nacional / MCTI, The Ohio State University, Pennsylvania State University, Shanghai Astronomical Observatory, United Kingdom Participation Group, Universidad Nacional Autónoma de México, University of Arizona, University of Colorado Boulder, University of Oxford, University of Portsmouth, University of Utah, University of Virginia, University of Washington, University of Wisconsin, Vanderbilt University, and Yale University. This publication uses Galaxy Zoo classifications; we thank the volunteers who made them possible. We used the MPA-JHU catalogues, the Simard et al. (2011) VizieR catalogue, and Astropy (Astropy Collaboration et al. 2022), Astroquery (Ginsburg et al. 2019), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), pandas (McKinney 2010), Matplotlib (Hunter 2007), statsmodels (Seabold & Perktold 2010), and scikit-learn (Pedregosa et al. 2011).

References

- Ahumada, R., Allende Prieto, C., Almeida, A., et al. 2020, *ApJS*, 249, 3
- Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, *ApJ*, 935, 167
- Bamford, S. P., Nichol, R. C., Baldry, I. K., et al. 2009, *MNRAS*, 393, 1324
- Bitsakis, T., Charmandaris, V., da Cunha, E., et al. 2011, *A&A*, 533, A142
- Blanton, M. R., Dalcanton, J., Eisenstein, D., et al. 2001, *AJ*, 121, 2358
- Brinchmann, J., Charlot, S., White, S. D. M., et al. 2004, *MNRAS*, 351, 1151
- Coenda, V., Muriel, H., & Martinez, H. J. 2012, *A&A*, 543, A119
- Dariush, A. A., Raychaudhury, S., Ponman, T. J., et al. 2010, *MNRAS*, 405, 1873
- Díaz-Giménez, E. & Mamon, G. A. 2010, *MNRAS*, 409, 1227
- Díaz-Giménez, E., Taverna, A., Zandivarez, A., & Mamon, G. A. 2020, *MNRAS*, 492, 2588
- Díaz-Giménez, E., Zandivarez, A., & Taverna, A. 2018, *A&A*, 618, A157
- Domínguez Sánchez, H., Huertas-Company, M., Bernardi, M., Tucillo, D., & Fischer, J. L. 2018, *MNRAS*, 476, 3661

- Dressler, A. 1980, *ApJ*, 236, 351
- 740 Ginsburg, A., Sipőcz, B. M., Brasseur, C. E., et al. 2019, *AJ*, 157, 98
- Graham, A. W., Driver, S. P., Petrosian, V., et al. 2005, *AJ*, 130, 1535
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Nature*, 585, 357
- Hickson, P. 1982, *ApJ*, 255, 382
- Hickson, P., Kindl, E., & Auman, J. R. 1989, *ApJS*, 70, 687
- Hickson, P., Mendes de Oliveira, C., Huchra, J. P., & Palumbo, G. G. 1992, *ApJ*, 399, 353
- Hunter, J. D. 2007, *Computing in Science and Engineering*, 9, 90
- Johnson, K. E., Hibbard, J. E., Gallagher, S. C., et al. 2007, *AJ*, 134, 1522
- 750 Jones, L. R., Ponman, T. J., Horton, A., et al. 2003, *MNRAS*, 343, 627
- Kauffmann, G., Heckman, T. M., Tremonti, C., et al. 2003a, *MNRAS*, 346, 1055
- Kauffmann, G., Heckman, T. M., White, S. D. M., et al. 2003b, *MNRAS*, 341, 33
- Kewley, L. J., Dopita, M. A., Sutherland, R. S., Heisler, C. A., & Trevena, J. 2001, *ApJ*, 556, 121
- Lim, S. H., Mo, H. J., Lu, Y., Wang, H., & Yang, X. 2017, *MNRAS*, 470, 2982
- 760 Lintott, C., Schawinski, K., Bamford, S., et al. 2011, *MNRAS*, 410, 166
- Lintott, C. J., Schawinski, K., Slosar, A., et al. 2008, *MNRAS*, 389, 1179
- Mahajan, S., Mamon, G. A., & Raychaudhury, S. 2011, *MNRAS*, 416, 2882
- Mamon, G. A. 1986, *ApJ*, 307, 426
- Mamon, G. A. 1992, *ApJ*, 401, L3
- McKinney, W. 2010, in *Proceedings of the 9th Python in Science Conference*, 56–61
- 770 McLachlan, G. & Peel, D. 2000, *Finite Mixture Models* (New York: Wiley)
- Mendes de Oliveira, C. & Hickson, P. 1994, *ApJ*, 427, 684
- Montaguth, G. P., Mendes de Oliveira, C., Lima-Dias, C., et al. 2026a, *ApJ*, 999, 160
- Montaguth, G. P., Monachesi, A., Torres-Flores, S., et al. 2025, *A&A*, 696, A240
- Montaguth, G. P., O’Mill, A. L., Mendes de Oliveira, C., et al. 2026b, *ApJ*, 998, 91
- 780 Montaguth, G. P., Torres-Flores, S., Monachesi, A., et al. 2023, *MNRAS*, 524, 5340
- Oman, K. A. & Hudson, M. J. 2016, *MNRAS*, 463, 3083
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. 2011, *J. Mach. Learn. Res.*, 12, 2825
- Peng, Y., Maiolino, R., & Cochrane, R. 2015, *Nature*, 521, 192
- Planck Collaboration, Ade, P. A. R., Aghanim, N., et al. 2016, *A&A*, 594, A13
- Ponman, T. J., Allan, D. J., Jones, L. R., et al. 1994, *Nature*, 369, 462
- 790 Salim, S., Rich, R. M., Charlot, S., et al. 2007, *ApJS*, 173, 267
- Seabold, S. & Perktold, J. 2010, in *Proceedings of the 9th Python in Science Conference*, 92–96
- Simard, L., Mendel, J. T., Patton, D. R., Ellison, S. L., & McConnell, A. W. 2011, *ApJS*, 196, 11
- Taverna, A., Salerno, J. M., Daza-Perilla, I. V., et al. 2023, *MNRAS*, 520, 6367
- Tempel, E., Tuvikene, T., Kipper, R., & Libeskind, N. I. 2017, *A&A*, 602, A100
- 800 Tricottet, M., Mamon, G. A., & Díaz-Giménez, E. 2025, *A&A*, 699, A329
- Tzanavaris, P., Hornschemeier, A. E., Gallagher, S. C., et al. 2010, *ApJ*, 716, 556
- Verdes-Montenegro, L., Yun, M. S., Williams, B. A., et al. 2001, *A&A*, 377, 812
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, *Nature Methods*, 17, 261
- Wainer, H. & Thissen, D. 1976, *Psychometrika*, 41, 9
- Wetzel, A. R., Tinker, J. L., Conroy, C., & van den Bosch, F. C. 2013, *MNRAS*, 432, 336
- 810 Zepf, S. E. & Whitmore, B. C. 1993, *ApJ*, 418, 72
- Zheng, Y.-L. & Shen, S.-Y. 2021, *ApJ*, 911, 105

Appendix A: SDSS data query

For the SDSS reference sample, we extract `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from `galSpecExtra`, `h_alpha_eqw` and the `[N II]`, `H $\alpha$` , `[O III]`, and `H $\beta$` fluxes from `galSpecLine`, and `p_el_debiased` and `p_cs_debiased` from `zooSpec`. In that query, these spectroscopic tables are joined to `SpecObj` by `specObjID`, while the photometric object is linked through `SpecObj.bestObjID=PhotoObj.objID`. The full result is split into the AGN-inclusive table and the non-AGN reference used for GMM calibration. The group samples instead inherit their DR12 `sfr_tot_p50`, `lgm_tot_p50`, and stored `specObjID` from the parent catalogues; the Galaxy Zoo vote fractions and the emission-line quantities are added to those rows by photometric `objid`, and the `galSpecIndx` indices `d4000_n` and `lick_hd_a` (with errors; sentinel values and non-positive errors treated as missing) by the stored DR12 `specObjID`. Their final `log sSFR` is the stored difference `sfr_tot_p50-lgm_tot_p50`.

We audited all 6076 final sample rows against `SpecObjAll` in DR12, retaining the stored `specObjID` even when the same photometric `objid` has several spectra (looking these DR12 identifiers up in DR16 fails for reprocessed BOSS observations, whose identifier encodes the reduction version). Of the 394 rows without a valid `sSFR`, 332 are BOSS-instrument spectra (`survey = boss`, `run2d = v5_7_0`) without any `galSpecExtra` or `galSpecInfo` row, because the MPA-JHU tables cover the legacy `run2d = 26` reductions only; 60 are legacy spectra with a `galSpecExtra` row, a valid stellar mass, `reliable = 1`, and `zWarning = 0`, whose `sfr_tot_p50` is the `-9999` sentinel; and 2 (both in `Control4B`) have no stored spectrum identifier. Per sample the BOSS/sentinel counts are 14/4 in `CG4`, 149/29 in `Control4B`, 164/27 in `Control4C`, and 5/0 in `RG4`. The missing rows are not low-signal-to-noise spectra: their median `snMedian_r` is 45 against 39 for the classified rows (rank-sum $p < 10^{-4}$ over unique spectra), and the same holds within each sample. The missing class is therefore a coverage gap of the MPA-JHU value-added catalogues plus a small number of failed SFR fits, unrelated to the star-formation properties of the galaxies; its extreme-assignment bounds are given in Appendix B.

Appendix B: sSFR classification details

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\text{N II}]\lambda 6584/\text{H}\alpha) \quad \text{and} \quad \log_{10}([\text{O III}]\lambda 5007/\text{H}\beta).$$

Among these galaxies, we flag AGN-like systems if they satisfy either

$$\log_{10} \frac{[\text{N II}]}{\text{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of Kauffmann et al. (2003a):

$$\log_{10} \frac{[\text{O III}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{N II}]/\text{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the GMM calibration sample. This calibration flag is deliberately more inclusive

Table B.1. Extreme-assignment bounds on the raw `CG4-minus-control` quenched-fraction difference. “Missing” columns give the missing-sSFR percentage of each side; the observed difference uses classified galaxies only; min/max are the extreme corners with all missing galaxies on each side assigned to quenched or star-forming (all-galaxy denominators).

Scope	Control	Missing (%)		Δf_Q^{obs}	min	max
		<code>CG₄</code>	ctrl			
All	<code>Control_{4B}</code>	7.3	6.4	0.025	-0.046	0.091
All	<code>Control_{4C}</code>	7.3	6.8	0.040	-0.034	0.107
All	<code>RG₄</code>	7.3	2.2	0.174	0.115	0.210
Sat.	<code>Control_{4B}</code>	6.5	5.8	0.043	-0.021	0.101
Sat.	<code>Control_{4C}</code>	6.5	6.3	0.063	-0.005	0.122
Sat.	<code>RG₄</code>	6.5	1.2	0.201	0.156	0.233

than the three-class BPT scheme of Sect. 3.5, in which composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations and only galaxies above the Kewley et al. (2001) line are labelled AGN-like, because its sole purpose is to keep possible AGN contamination out of the sSFR calibration. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN; they are retained for the sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available, while the AGN-like analysis treats them separately. Reference-sample rows with unavailable `specsfr_tot_p50`, and group-sample rows with unavailable `sfr_tot_p50` or `lgm_tot_p50`, including the `-9999` sentinel, have no valid sSFR estimate; they are treated as missing data throughout: excluded from the GMM calibration where applicable, never assigned an sSFR class, omitted from every sSFR figure, and reported as counts only (Sect. 2.2). Galaxies with a measured sSFR are divided into quenched and star-forming classes with a two-component Gaussian Mixture Model (GMM) in the $\log M_*$ - $\log$ sSFR plane (McLachlan & Peel 2000); the boundary is the locus where the two component posterior probabilities are equal, and the resulting classification for the SDSS non-AGN reference is shown in Fig. B.1. A fixed-threshold check confirms that the qualitative hierarchy is not tied to the exact GMM split (Sect. 3.2).

Extreme-assignment bounds for the missing class. Because the missing-sSFR class could in principle hide differential quenching, Table B.1 bounds every raw quenched-fraction contrast by the four extreme assignments in which *all* missing-sSFR galaxies on *each* side are set to quenched or to star-forming, with consistent all-galaxy denominators on both sides. The differential missingness is negligible against `Control4B` and `Control4C` (7% of `CG4` members versus 6% and 7%) and material only against `RG4` (2%). The raw `RG4` contrast survives all four corners (all-member difference between 0.12 and 0.21 around the observed 0.17; satellites 0.16 to 0.23), whereas the small `Control4B` and `Control4C` differences can change sign only under the doubly adversarial corner and are not the basis of any claim in this paper.

Star formation at fixed morphology. Table B.2 gives the quenched fractions among sSFR-classified galaxies at fixed vote-based morphology that enter the homogeneity test and the Kitagawa decomposition of Sect. 3.2; the decomposition splits each raw `CG4-minus-control` quenched-fraction offset

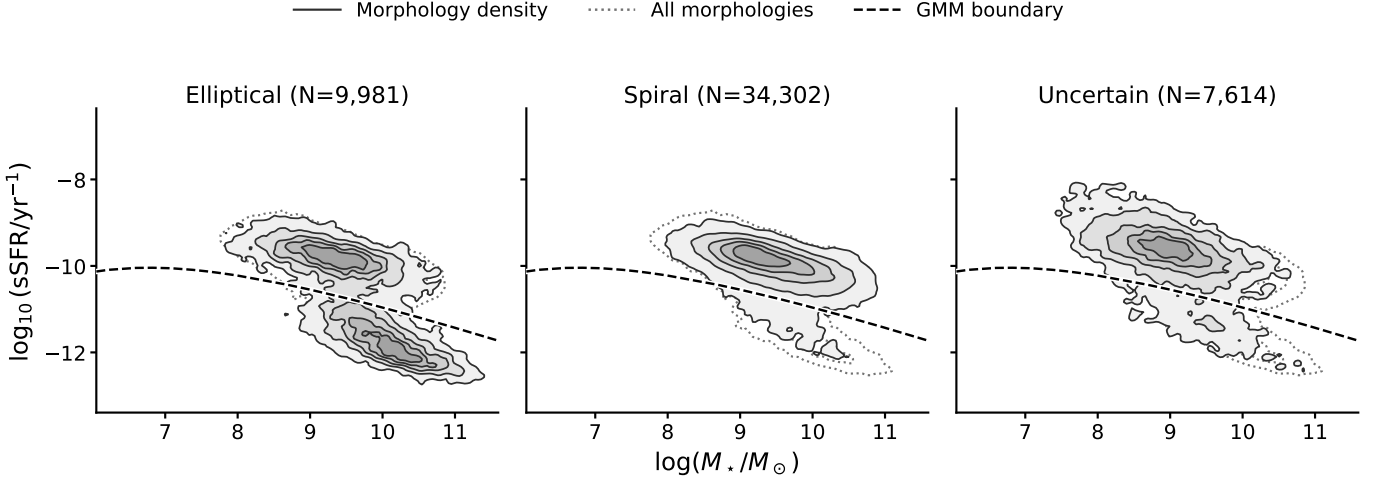


Fig. B.1. Galaxy Zoo morphology in sSFR–stellar-mass space for the SDSS non-AGN reference, excluding galaxies without sSFR estimates. Panels show elliptical, spiral, and uncertain morphologies; no unavailable morphologies occur in this reference sample ($N = 0$). Solid class-specific contours enclose 20, 40, 60, 80, and 95% of the density, grey dotted contours show the full reference distribution at the same enclosed-probability levels, and the dashed curve is the adopted star-forming–quenched GMM boundary.

Table B.2. Conditional quenched fractions $P(Q|\text{morphology})$ among sSFR-classified galaxies, with Wilson 95% intervals.

	CG ₄	Control _{4B}	Control _{4C}	RG ₄
Elliptical	0.908 ^{+0.040} _{-0.065}	0.903 ^{+0.016} _{-0.019}	0.887 ^{+0.017} _{-0.020}	0.850 ^{+0.069} _{-0.111}
Spiral	0.326 ^{+0.105} _{-0.090}	0.357 ^{+0.027} _{-0.026}	0.328 ^{+0.027} _{-0.026}	0.295 ^{+0.084} _{-0.072}

Notes. For the uncertain-morphology class, $P(Q|U) = 0.429, 0.552, 0.594$, and 0.267 ($n = 7, 116, 143$, and 15 classified galaxies, same sample order).

Table C.1. CG₄ galaxies inside would-be Control_{4C} quartets of the parent catalogue, by compact-group class.

CG ₄ class	CG ₄ groups	Lim groups	Galaxies
Embedded	14 / 19	14	52
Isolated	6 / 6	6	24
Predominant	37 / 37	37	145
Split	4 / 16	4	12

Notes. “CG₄ groups” counts compact groups with at least one member inside a parent-group quartet, over the total number of compact groups of that class (split groups included, as in the exclusion).

Appendix C: Compact groups as projected cores of ordinary groups

The compact-group exclusion of Sect. 2.1 is not a technicality: for 61 of the 765 parent Lim groups, the would-be Control_{4C} quartet (BGG plus the three closest projected companions within 3 magnitudes) contains at least one CG₄ galaxy—233 CG₄ galaxies in total. Table C.1 breaks this overlap down by the Zheng & Shen (2021) class: 37 of the 37 predominant and 14 of the 19 embedded compact groups are the projected cores of their host groups, as Fig. 1 illustrates; the remaining embedded systems sit in rich hosts whose BGG-centred core is made of other members. The isolated entries are trivial self-overlap: for each of the 6 isolated systems, the overlapping Lim group is the compact group’s own host, whose BGG and three nearest projected companions are exactly the four CG₄ members. Had these groups not been excluded, the same galaxies would have appeared on both sides of the comparisons, biasing every contrast toward null.

Appendix D: Statistical conventions

Independence unit. The unit for every inferential contrast among the group catalogues is the *physical* Lim group: standard errors are clustered by Lim group (unifying the over-

lapping control definitions, with CG₄ systems assigned to their host Lim group, or to their own identifier when they have no Lim counterpart), and resampling of grouped contrasts is blocked by physical group or by matched group/-pair block. Pooled models across the three controls are secondary summaries computed after deduplicating the overlapping control samples to one row per physical galaxy, with the kept label following the priority RG₄ > Control_{4B} > Control_{4C} (Appendix H). Purely descriptive reference-sample diagnostics report their own unit where they appear.

Tests and p-values. Descriptive comparisons use two-sided Fisher exact tests for 2×2 tables and rank-sum or bootstrap comparisons for distributions and median differences; Spearman and Pearson coefficients are diagnostics of monotonic or linear trends only. Paired sign-flip permutation p-values use add-one tail counts, $p = (k + 1)/(B + 1)$ with k the number of permutations at least as extreme as the observation; bootstrap sign-crossing p-values are $2 \min[(k_{\leq 0} + 1)/(B + 1), (k_{\geq 0} + 1)/(B + 1)]$, capped at unity, where $k_{\leq 0}$ and $k_{\geq 0}$ count resampled differences on either side of zero; displayed p-values are floored at 10^{-4} . Adjusted odds ratios come from binomial generalised linear models with cluster-robust covariance; continuous covariates are stan-

dardised, so coefficients of continuous terms are per standard deviation of the fitted frame. Multiplicity control is reserved for the exploratory batteries: Holm adjustment for the colour, measurement-availability, and magnitude-gap screens, and Benjamini–Hochberg false-discovery-rate control for the dominance, morphology–environment, and second-order screens; the Holm values across the three per-control matched contrasts (Table 4) are bookkeeping, since the controls are separate pre-specified comparisons.

Covariates. The velocity dispersion σ_v that enters models and matching is the quartet-level gapper estimate (Wainer & Thissen 1976) on the four selected members’ redshifts, converted as $c\sigma_z/(1+\bar{z})$, for *all* samples, including Control_{4B} and Control_{4C} quartets embedded in richer hosts (this is how the spurious Lim 3688 quartets of Sect. 2.1 reach $\approx 2000 \text{ km s}^{-1}$). With four tracers the estimator is intrinsically noisy: in a Gaussian Monte Carlo of 10^5 quartets, $\sigma_{\text{gapper}}/\sigma_{\text{true}}$ has median 0.96 with a 16–84% range of 0.57–1.42. A candidate covariate enters an adjusted model when it is non-missing for at least 65% of that model’s frame (matching variables require 70%); in practice σ_v , group luminosity, stellar mass, redshift, and rank are complete for every quartet and always enter, and no harmonised group-mass covariate is available (Appendix H.3).

Overlap coefficient. Where two distributions are compared for common support (compactness in Sect. 2.1, $\log T_i$ in Sect. 3.4), the overlap coefficient is the integral of the minimum of the two kernel density estimates, between 0 (disjoint) and 1 (identical).

Appendix E: Main-sequence and quenched-sequence details

We model the star-forming main sequence as a function of stellar mass with polynomials of order 1–4 fitted to the star-forming galaxies of the SDSS non-AGN reference sample; the diagnostic is the residual RMS scatter, not a formal reduced χ^2 , because no per-galaxy uncertainty model is used. We adopt order 2, the lowest order with the smallest five-fold cross-validated RMS (0.305 dex, against 0.306 dex for order 1), and compute the residual of each star-forming galaxy with respect to it; a positive residual means a higher sSFR at fixed stellar mass than the reference relation. In Sect. 3.2, Δ is the CG₄-minus-control median residual with the central 68% group-bootstrap interval on the median difference and the two-sided sign-crossing probability (10 000 resamples). The quenched sequence of Fig. 4b is obtained with the same procedure on the 7705 quenched galaxies of the reference sample: order 4 has the smallest cross-validated RMS (0.331 dex), and the residuals are attached to the quenched galaxies of the four samples.

Within CG₄, the star-forming main-sequence offsets of the embedded, isolated, and predominant classes are statistically consistent with one another.

Appendix F: Secondary and environmental diagnostics

This appendix collects the secondary diagnostics summarised in Sect. 3.5.

F.1. Magnitude gaps

We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap; large gaps are the classic optical signature of fossil-like assembly (Ponman et al. 1994; Jones et al. 2003; Dariush et al. 2010).

Under the Holm adjustment of the magnitude-gap battery, larger gaps appear only against Control_{4B}: for Δm_{12} , the CG₄ and Control_{4B} medians are 1.17 and 0.85 mag (adjusted $p < 10^{-4}$), and for Δm_{14} they are 2.56 and 1.90 mag (adjusted $p < 10^{-4}$); the Control_{4C} and RG₄ differences are weak for both gap definitions.

A deduplicated pooled sensitivity gives control medians of 0.86 mag for Δm_{12} and 1.91 mag for Δm_{14} (adjusted $p < 10^{-4}$ and $p < 10^{-4}$); because the deduplicated control side is dominated by Control_{4B}-selected systems, it is not evidence for a control-robust gap excess, and the morphology signal is not accompanied by a comparably robust excess in these fossil-like indicators.

In the 14-test gap-correlation screen over all 1519 CG₄ and control quartets (control quartets entering once per label, without deduplication), the relation between Δm_{12} and the quenched satellite fraction has $\rho_S = -0.18$ (adjusted $p < 10^{-4}$).

F.2. Projected phase space

We ask whether the CG₄–RG₄ satellite contrasts vary with projected position relative to the BGG, using 174 CG₄ and 166 RG₄ satellites ranked by projected BGG-centric distance and, when available, $|\Delta v_{\text{BGG}}|/\sigma_v$; such coordinates are statistical proxies for infall state, not orbital reconstructions (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the quenched fractions are 68.3% for CG₄ and 48.2% for RG₄, a difference of 20.1 percentage points; the corresponding elliptical fractions are 64.7% and 35.6% (difference 29.1 points).

The quenched-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG₄ offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the quenched-status model gives a CG₄ odds ratio of 1.18 (95% CI 0.43–3.24; $p = 0.742$), and the analogous elliptical model gives 2.42 ($p = 0.078$).

A nearest-neighbour mass–redshift matched check on 126 CG₄–RG₄ satellite pairs gives matched quenched- and elliptical-fraction differences of 9.5 and 18.7 percentage points.

In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites, so these bins locate where the contrast appears; they are not a dynamical clock.

F.3. Distance to the BGG

We compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

The following sample-specific monotonic trends have small p-values.

In CG₄, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.24$ for 174 satellites ($p = 1.5 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 5.5 \times 10^{-3}$).

In Control_{4B}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.08$ for 1972 satellites ($p = 3.2 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 1.9 \times 10^{-4}$).

In Control_{4C}, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.19$ for 1977 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.16$ ($p < 10^{-4}$).

In Control_{4C}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.11$ for 1977 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.11$ ($p < 10^{-4}$).

F.4. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity, and non-dominated systems. Because this appendix screens many group properties, p_{adj} denotes the false-discovery-rate adjustment of Appendix D; Cliff's δ is the rank-based effect size. The split does not produce a compact-group signal by itself: across the control samples, the clearest binary separation is in group luminosity, dominated systems having lower L_{group} than their non-dominated counterparts (Control_{4B}: 9.1×10^{10} versus $1.28 \times 10^{11} L_{\odot}$, $p_{\text{adj}} < 10^{-4}$, $\delta = -0.55$; Control_{4C}: 9.66×10^{10} versus $1.07 \times 10^{11} L_{\odot}$, $p_{\text{adj}} < 10^{-4}$, $\delta = -0.25$; RG₄: 7.09×10^{10} versus $9.82 \times 10^{10} L_{\odot}$, $p_{\text{adj}} = 1 \times 10^{-4}$, $\delta = -0.59$), the negative δ meaning that dominated groups occupy the lower-luminosity side of each comparison.

Splitting instead at each sample's median BGG luminosity fraction ($f_{L,\text{BGG}} = 0.63, 0.52, 0.61$, and 0.59 for CG₄, Control_{4B}, Control_{4C}, and RG₄) gives the same picture: the FDR-significant separations are Control_{4B} in L_{group} ($L_{\odot}$) ($\delta = -0.53$, $p_{\text{adj}} = 0.000$), Control_{4C} in L_{group} ($L_{\odot}$) ($\delta = -0.24$, $p_{\text{adj}} = 0.000$), RG₄ in L_{group} ($L_{\odot}$) ($\delta = -0.58$, $p_{\text{adj}} = 0.000$), Control_{4B} in $\Delta_{\text{BGG-cen}}/\langle R_{ij} \rangle$ ($\delta = -0.17$, $p_{\text{adj}} = 0.000$), Control_{4B} in σ_v (km s⁻¹) ($\delta = -0.12$, $p_{\text{adj}} = 0.024$), Control_{4C} in $\Delta_{\text{BGG-cen}}/\langle R_{ij} \rangle$ ($\delta = -0.11$, $p_{\text{adj}} = 0.028$), and none involves CG₄.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in Control_{4C}, where S_{frac} increases with $\langle R_{ij} \rangle$ in both dominated and non-dominated groups ($\rho_S = 0.30$, $p < 10^{-4}$ and $\rho_S = 0.26$, $p < 10^{-4}$), while weaker negative trends are found with σ_v and L_{group} . Control_{4B} contributes only weaker non-dominated-group correlations. In CG₄, the reported trends occur in small subsets, with S_{frac} decreasing with L_{group} among dominated groups ($\rho_S = -0.40$, 36 groups) and with $\Delta V_{\text{BGG}}/\sigma_v$ among non-dominated groups ($\rho_S = -0.45$, 24 groups); we treat these as supporting diagnostics rather than the main dominance result. The full distributional diagnostics are provided in the online supplement.

F.5. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, the following cases have small p-values.

- Control_{4B} dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.842 ($p = 0.01$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.842 ($p = 0.02$).
- Control_{4C} dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.972 ($p < 10^{-4}$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.972 ($p < 10^{-4}$).

In each case the group-level and satellite-level clustered treatments share the same point estimate by construction and differ only in the uncertainty model, which is why the odds ratios coincide.

F.6. Morphology versus Zheng–Shen class, magnitude gap, and BGG dominance

We test whether the E/S morphology mix is associated with three group-level descriptors: the Zheng–Shen compact-group class, the first–second magnitude gap $\Delta m_{12} = M_{r,2} - M_{r,1}$, and the continuous BGG luminosity fraction $f_{L,\text{BGG}} = L_{\text{BGG}}/L_{\text{group}}$. These tests use only the elliptical and spiral classes; uncertain and missing morphologies are excluded. We run the tests separately for all galaxies, BGGs only, and satellites only, so that any trend driven by the BGG itself is not folded into the satellite result; for satellites, $f_{L,\text{BGG}}$ mechanically includes the satellite luminosity in the group-luminosity denominator, so the satellite models carry a stellar-mass control. The magnitude-gap and BGG-luminosity-fraction quantities are available for all 118 CG₄+RG₄ groups used in this test; the Zheng–Shen class is defined only for compact groups, so the RG₄ class test is unavailable while RG₄ is retained in the Δm_{12} and $f_{L,\text{BGG}}$ comparisons. Overall these tests provide little evidence that the E/S mix is driven by class, magnitude gap, or BGG luminosity fraction (Tables F.1 and F.3); the class structure of the elliptical fraction is summarised in Table F.2 and Fig. 5, with group-level permutation tests on the per-group mean satellite f_E (20 000 permutations, two-sided, add-one convention) giving $p = 0.023$ for isolated versus the rest of CG₄, $p = 1.00$ for isolated versus RG₄, and $p = 0.40$ for isolated versus Control_{4C}.

The Δm_{12} , $f_{L,\text{BGG}}$, and CG₄–RG₄ interaction coefficients are all weak across the all-galaxy, BGG, and satellite fits; a class interaction is undefined because the Zheng–Shen class has no RG₄ counterpart, and no result survives the exploratory FDR screen.

F.7. Crossing time

Group radii and t_{cross} use projected proper separations from the Planck 2015 angular-diameter distance, correcting the luminosity-distance convention used for this diagnostic in

Table F.1. Elliptical and spiral counts by Zheng–Shen class for the CG₄ sample. Uncertain morphologies and rows without Galaxy Zoo vote fractions are excluded; split systems are absent from the non-split CG₄ sample.

Zheng–Shen class	All CG ₄ E/S	CG ₄ BGG E/S	CG ₄ sat. E/S
Split	0/0	0/0	0/0
Isolated	7/11	3/1	4/10
Predominant	76/53	22/11	54/42
Embedded	41/22	11/5	30/17

Table F.2. Elliptical fractions $f_E = N_E/(N_E + N_S)$ and median host halo mass by Zheng–Shen class, with Wilson 95% intervals.

Class	Groups	f_E (all)	f_E (sat.)	$\log M_{200c}$
Isolated	6	$0.389^{+0.225}_{-0.186}$	$0.286^{+0.261}_{-0.169}$	12.86
Embedded	19	$0.651^{+0.106}_{-0.123}$	$0.638^{+0.122}_{-0.143}$	13.83
Predominant	37	$0.589^{+0.081}_{-0.086}$	$0.562^{+0.095}_{-0.100}$	13.13

Notes. Host $\log M_{200c}$ is the Yang–Lim abundance-matching host-halo mass (Appendix H.3); for isolated systems it describes the quartet itself. Embedded compact groups contribute less than half of their host luminosity and therefore live in the richer, more massive hosts.

Table F.3. Tests of the E-vs-S morphology mix against compact-group class, magnitude gap, and BGG luminosity fraction. The continuous columns report the available adjusted logistic model for $P(E)$; all-galaxy and satellite fits use group-clustered standard errors. Odds ratios are per magnitude for Δm_{12} and per 0.1 increase in luminosity fraction for $f_{L,BGG}$.

Subset	Sample	Test	Result
all galaxies	CG ₄	Class	$p = 0.137$
		Δm_{12}	OR 1.42, $p = 0.167$
		$f_{L,BGG}$	OR 1.29, $p = 0.080$
all galaxies	RG ₄	Class	not defined
		Δm_{12}	OR 1.33, $p = 0.220$
		$f_{L,BGG}$	OR 1.09, $p = 0.540$
BGGs only	CG ₄	Class	$p = 1.000$
		Δm_{12}	OR 1.77, $p = 0.426$
		$f_{L,BGG}$	OR 1.68, $p = 0.174$
BGGs only	RG ₄	Class	not defined
		Δm_{12}	OR 1.58, $p = 0.312$
		$f_{L,BGG}$	OR 1.21, $p = 0.520$
satellites only	CG ₄	Class	$p = 0.066$
		Δm_{12}	OR 1.28, $p = 0.346$
		$f_{L,BGG}$	OR 1.18, $p = 0.321$
satellites only	RG ₄	Class	not defined
		Δm_{12}	OR 1.41, $p = 0.342$
		$f_{L,BGG}$	OR 1.12, $p = 0.584$

Tricottet et al. (2025); the values decrease by 3.3–9.6% (median 7.2%), without changing the reported correlations at quoted precision. The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity: shorter t_{cross} systems have larger virial mass-to-light ratios and virial masses, with correlations of $\rho_S = -0.39$ ($p < 10^{-4}$) with $\log(M_{\text{VT}}/L_r)$ and $\rho_S = -0.33$ ($p < 10^{-4}$) with $\log(M_{\text{VT}})$ over 816 groups, whereas the relation with total group luminosity is formally detected but very weak ($\rho_S = 0.08$). We therefore read t_{cross} as a dynamical compactness indicator tied to virial mass and mass-to-light ratio, with little independent leverage from L_{group} .

F.8. Optical colours

We derive the $(u-r)$, $(u-g)$, $(g-r)$ and $(r-i)$ colours from the matched SDSS photometry and model each colour jointly with $\log(M_*/M_\odot)$ and redshift. The retained colour-analysis matches include 110 of 248 galaxies in CG₄, 931 of 2792 galaxies in Control_{4B}, 1153 of 2812 galaxies in Control_{4C}, 102 of 224 galaxies in RG₄. This stricter subset differs from the completeness of Fig. H.3, whose colour row only requires the four derived SDSS colour columns to be present in the final catalogue (217 of 248 CG₄ galaxies), whereas the colour analysis also requires successful matched-photometry retention and the covariates used below; the fractions retained are 44.4% in CG₄, 33.3% in Control_{4B}, 41.0% in Control_{4C}, 45.5% in RG₄. The colour subset is not representative of the full catalogues: matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies.

The raw catalogue colour–mass fits indicate that CG₄ does not simply follow the same colour–mass relation as every comparison sample: common slopes are rejected in all four colours, with the directional slope differences having small adjusted p-values mainly in the bluer colours and depending on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. We refit satellite colours with cluster-robust standard errors: in the pooled mass- and redshift-adjusted comparison, the negative CG₄ coefficients are weak, and adding morphology and sSFR class does not reveal a clear negative coefficient. These coefficients remain control-sample dependent, with non-uniform split-sample behaviour, so they do not identify a global colour shift or a simple interaction-triggered star-formation population (Fig. F.1). Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre, and redder BGGs tend to have redder mean satellite populations in the combined group sample, but both use the same biased colour subset and are exploratory; the CG₄ satellite blue-fraction correlation with crossing time, $\rho_S = 0.365$ (adjusted $p = 0.050$), is a boundary result that does not change this reading. Detailed colour–mass, satellite, and radial diagnostics are provided in the online supplement.

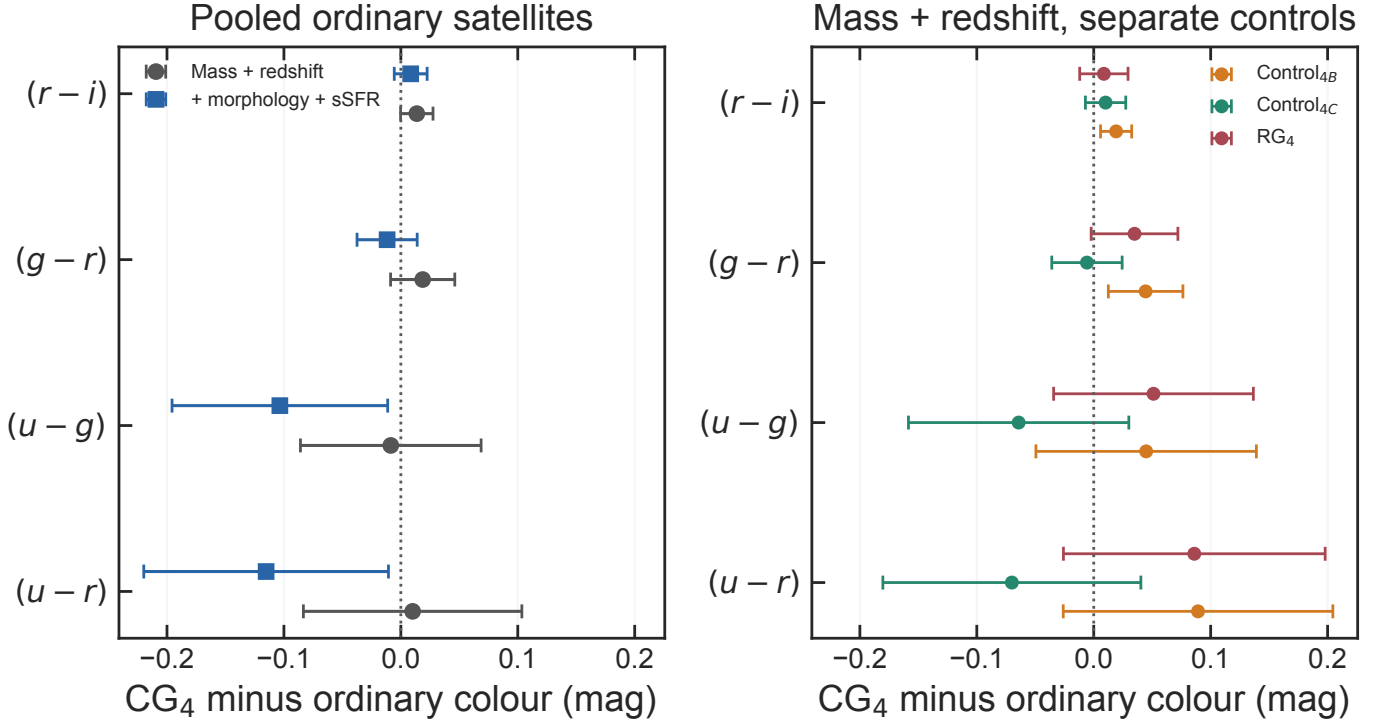


Fig. F.1. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG_4 minus ordinary coefficient and bars show 95% confidence intervals; negative values indicate bluer CG_4 satellites.

Appendix G: Second-order descriptors of the satellite morphology mix

As an exploratory complement to the class comparison of Sect. 3.3, Table G.1 fits the elliptical-versus-spiral classification of the satellites, adjusted for stellar mass and clustered by group, against five group-scale descriptors in turn: the median pairwise projected separation $\log R_{ij,med}$, the quartet velocity dispersion σ_v , the BGG luminosity fraction $f_{L,BGG}$, the crossing time $\log t_{cross}$, and an indicator for $f_{L,BGG}$ above the CG_4 median (0.63). Continuous descriptors are standardised, so their odds ratios are per standard deviation of the descriptor within each sample. The same models are fitted within each control for reference, and Benjamini–Hochberg control is applied across the 20 tests of the battery. Within CG_4 , only the compactness descriptor is associated with the satellite mix: more compact quartets have more elliptical satellites (odds ratio 0.65 per standard deviation of $\log R_{ij,med}$, 95% CI [0.50, 0.84], $p_{BH} = 0.005$), a trend shared by $Control_{4B}$ and $Control_{4C}$; σ_v , $f_{L,BGG}$, and t_{cross} show no association within CG_4 . The dominance tables of Appendix F.4 are also reproduced with a median $f_{L,BGG}$ split there.

Table G.1. Second-order battery: odds ratios of elliptical (versus spiral) satellite classification per descriptor, adjusted for stellar mass, with 95% intervals and Benjamini–Hochberg adjusted p-values across the whole battery. Continuous descriptors are per standard deviation within the sample; the last row is a binary indicator.

Descriptor	CG ₄	Control _{4B}	Control _{4C}	RG ₄
$\log R_{i,j,\text{med}}$	0.65 [0.50, 0.84] $p_{\text{BH}}=0.005$	0.80 [0.72, 0.89] $p_{\text{BH}}=1.2 \times 10^{-4}$	0.70 [0.63, 0.78] $p_{\text{BH}} < 10^{-4}$	0.92 [0.64, 1.34] $p_{\text{BH}}=0.76$
σ_v	0.90 [0.65, 1.25] $p_{\text{BH}}=0.74$	1.03 [0.93, 1.15] $p_{\text{BH}}=0.74$	1.21 [1.09, 1.35] $p_{\text{BH}}=0.002$	0.90 [0.61, 1.32] $p_{\text{BH}}=0.74$
$f_{L,\text{BGG}}$	1.19 [0.88, 1.59] $p_{\text{BH}}=0.59$	1.02 [0.90, 1.16] $p_{\text{BH}}=0.78$	1.06 [0.95, 1.19] $p_{\text{BH}}=0.59$	1.19 [0.79, 1.81] $p_{\text{BH}}=0.74$
$\log t_{\text{cross}}$	0.88 [0.66, 1.17] $p_{\text{BH}}=0.74$	0.87 [0.78, 0.97] $p_{\text{BH}}=0.03$	0.73 [0.65, 0.81] $p_{\text{BH}} < 10^{-4}$	0.99 [0.69, 1.43] $p_{\text{BH}}=0.98$
$f_{L,\text{BGG}} > \text{med}(\text{CG}_4)$	1.29 [0.64, 2.60] $p_{\text{BH}}=0.74$	1.24 [0.92, 1.67] $p_{\text{BH}}=0.44$	1.09 [0.87, 1.36] $p_{\text{BH}}=0.74$	1.19 [0.51, 2.77] $p_{\text{BH}}=0.76$

Notes. $N = 157$ CG₄ satellites in 59 groups; 1772, 1744, and 142 satellites in the controls.

Appendix H: Pooled models, matching details, and selection diagnostics

H.1. Pooled adjusted models

As a compact summary across the heterogeneous control definitions we also fit binomial models to the pooled catalogue. Because the control samples overlap heavily, the pooled control side is first deduplicated by SDSS objid and standard errors are clustered by physical Lim group; the common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity, and velocity dispersion. This pooled result does not replace the per-control contrasts. For quenched status, the CG₄ odds ratio is 1.35 (95% CI [0.98, 1.86]; $p = 0.068$). For quenched satellite status, the CG₄ odds ratio is 1.53 (95% CI [1.10, 2.11]; $p = 0.011$). For elliptical morphology, the CG₄ odds ratio is 1.77 (95% CI [1.32, 2.37]; $p = 1.4 \times 10^{-4}$). For elliptical satellite morphology, the CG₄ odds ratio is 2.08 (95% CI [1.51, 2.88]; $p < 10^{-4}$). For elliptical BGG morphology, the CG₄ odds ratio is 0.78 (95% CI [0.39, 1.53]; $p = 0.464$). The pooled coefficients reinforce the satellite morphology shift (Fig. H.1); the quenched estimate changes across the separate controls and is not reproduced by the matched contrast.

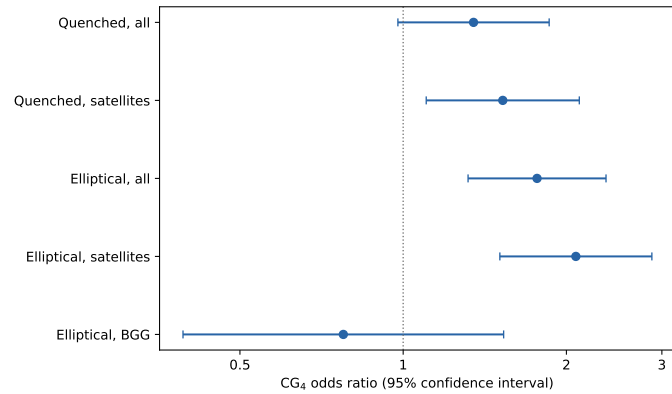


Fig. H.1. Adjusted CG₄ odds ratios for the star-formation and morphology outcomes in the pooled secondary models, including all-galaxy, satellite-only, and BGG-only rows. Bars show 95% confidence intervals; the vertical line marks equal odds.

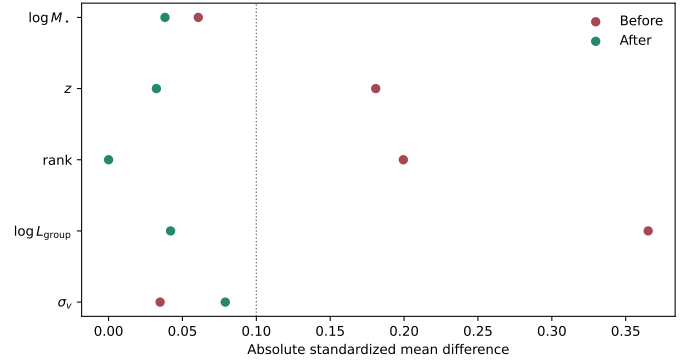


Fig. H.2. Absolute standardized mean differences before and after matching.

H.2. Galaxy-level matching details

The galaxy-level control pool of Sect. 3.1 is deduplicated by SDSS objid before matching: 5828 control rows collapse to 3609 unique galaxies. The propensity model uses $\log M_*$, redshift, rank, $\log L_{\text{group}}$, and σ_v , and the match is one-to-one without replacement within exact rank strata. The 14 unmatched CG₄ galaxies lay outside the common propensity-score support or exceeded the caliper. The maximum absolute covariate standardized mean difference falls from 0.37 to 0.08 (Fig. H.2). By kept label, the matched controls comprise 159 Control_{4B}, 57 Control_{4C}, 18 RG₄; physically, 18 of the 234 matched controls are RG₄ galaxies (which also appear in Control_{4B} through the sample overlap), and the per-galaxy origin of every matched control is released with the derived tables. Confidence intervals and p-values come from a bootstrap blocked by the treated galaxy’s compact group.

The 234 matched controls occupy 204 physical Lim groups, and 26 control hosts link more than one treated CG₄ group. Resampling the 33 connected components of the treated-group/control-host graph instead, the elliptical-fraction difference remains 0.139 (95% interval [0.051, 0.244]; $p = 0.004$), and the quenched-fraction difference is 0.031 ([−0.090, 0.120]; $p = 0.649$). The analogous matched-size sensitivities are −0.019 dex for the Simard radius ($p = 0.310$), −0.033 dex for the Petrosian radius ($p = 0.016$), and −0.008 for the concentration ($p = 0.896$).

H.3. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Fig. H.3, including the Simard and Petrosian size availability used in Sect. 3.5 and the $D_n4000/H\delta_A$ availability of the spectral indices. The median nearest-neighbour angular separation is 100.9 arcsec in CG₄ and 255.9 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 7.7%. No major availability difference has both an adjusted p-value below 0.05 and a difference exceeding 10 percentage points.

The group-scale covariates are radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. The tables do carry an M_{200c} estimate, but it is the Yang–Lim abundance-matching M_{180m} of the identified *host* halo (converted as in Paper I, Appendix A), not a quartet property: for embedded quartets it describes the richer host while for isolated systems it describes the quartet itself, it derives from group luminosity by abundance matching and is therefore nearly collinear with the group-luminosity covariate, and it is entangled with the compact/loose selection. We therefore exclude it from the harmonised covariate set. A sensitivity fit adding $\log M_{200c}$ to the published covariates leaves every elliptical conclusion unchanged: the pooled odds ratio moves from 1.77 to 1.62 ($p = 0.0020$), and the per-control odds ratios become 2.00, 1.56, and 2.65 against Control_{4B}, Control_{4C}, and RG₄ (all $p < 0.005$).

Restricting the pairwise tidal index of Sect. 3.4 to quartet co-members is innocuous: recomputing T_i with all spectroscopic Lim members of the identified host group included (positions from the group catalogue; stellar masses from MPA–JHU where available, so the host-inclusive index is itself a lower bound) shifts $\log T_i$ by a median of 0.009 dex (sample medians ≤ 0.02 dex, and identically zero for RG₄, whose groups have exactly four members), with rank correlation 0.91 between the two definitions, and re-fitting the with-index elliptical model on the host-inclusive index moves the attenuated CG₄ coefficient only from 1.39 to 1.33. T_i is thus the local, galaxy-scale tidal density traced by catalogued galaxies; any smooth host-halo contribution is probed separately through the host-mass sensitivity fit above and the within-host analysis of Sect. 3.1. The colour-selection mass diagnostics are provided in the online supplement.

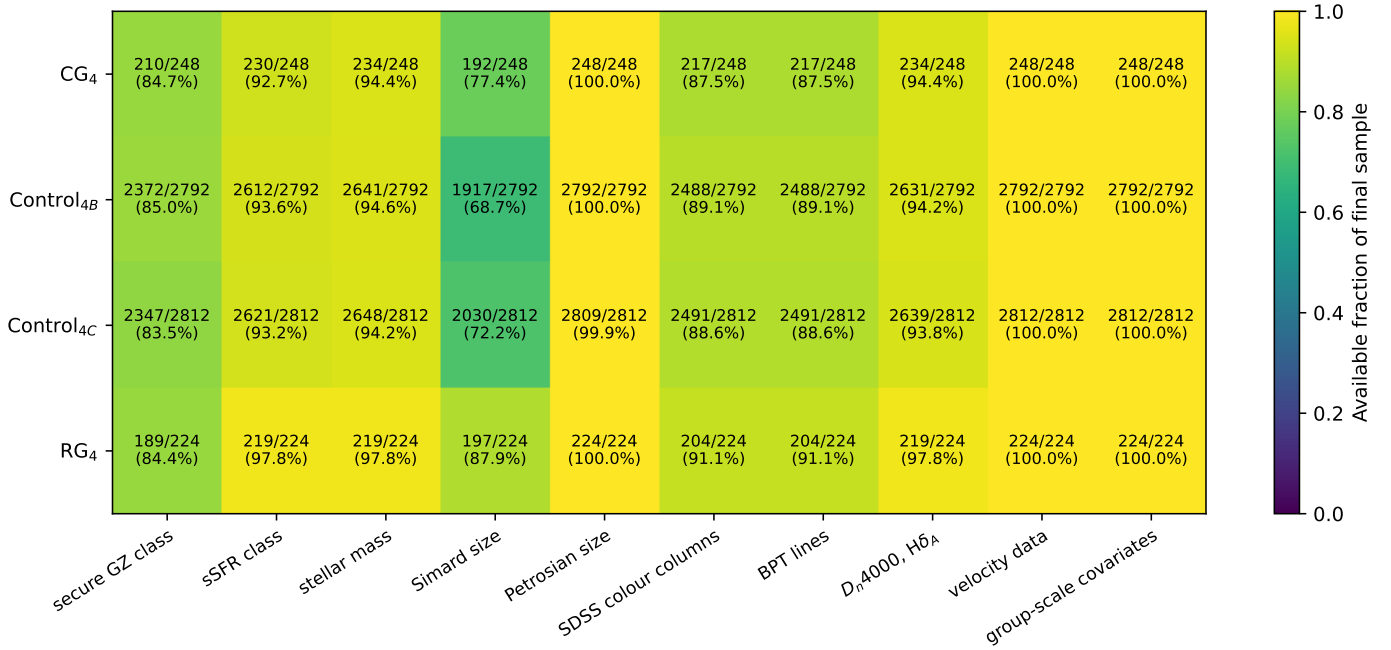


Fig. H.3. Availability fractions for the major derived quantities in each sample. Cell labels give the number available over the final per-sample galaxy count and the corresponding percentage. The colour row records completeness of the four derived SDSS colour columns in the final catalogue, rather than the stricter colour-analysis subset used in Appendix F.8; the D_n4000 , $H\delta_A$ column requires finite, non-sentinel `galSpecIndx` values with positive errors for the stored DR12 spectrum; the Simard-size column reflects the DR7 bridge, redshift, quality-window, blending, and pegged-Sérsic-index cuts described in Sect. 2.3.

Appendix I: Robustness details

Crowding. Galaxy Zoo morphology is image-based, so crowded closest pairs could bias the votes. After excluding galaxies whose nearest projected neighbour within the catalogue group lies within 55 arcsec (retaining 200 CG₄ and 2534 Control_{4C} galaxies), the CG₄ all-row elliptical fraction *rises* relative to the full-sample value of Table 3 (52.5% against 40.1% for Control_{4C}, all retained rows including uncertain and missing classes), and the CG₄–Control_{4C} contrast remains clear. In the joint model, the close-neighbour flag has OR 1.98 ($p = 6.1 \times 10^{-4}$), while the CG₄ coefficient at fixed flag status is OR 1.76 ($p = 2.6 \times 10^{-4}$); the former compares close with non-close galaxies, the latter CG₄ with controls at fixed flag status, so neither contradicts the rising raw CG₄ fraction after exclusion. The checks do not prove immunity to crowding, but they rule out the closest-pair artefact as the sole origin of the result.

Without σ_v . Refitting the per-control and pooled models without the σ_v covariate, on identical samples and with identical clustering, moves the elliptical odds ratios from 2.14 to 2.22 (Control_{4B}), 1.67 to 1.68 (Control_{4C}), and 2.81 to 2.76 (RG₄), less than their standard errors. The galaxy-level match retains the same 234 CG₄ galaxies with an elliptical-fraction difference of 0.16. The per-control group-level differences become $\Delta = 0.28, 0.10$, and 0.25 against Control_{4B}, Control_{4C}, and RG₄ (permutation $p = 0.0001, 0.056, 0.0016$). The Control_{4B} change reflects the composition of the matched set rather than the CG₄ side: the with- and without- σ_v matches share only 7 of 54 control groups, the matched-control mean elliptical-satellite fraction changes by -0.12 while the CG₄ mean is unchanged, and without σ_v the matched controls are dynamically cooler

than their CG₄ partners (residual σ_v standardized mean difference 0.27, medians 160 against 129 km s^{-1}) and less well balanced in redshift, so the with- σ_v match remains the primary one and the two values bracket the Control_{4B} contrast.

Within-host estimators. The conditional-logit sample of Sect. 3.1 contains 791 galaxies in 53 host strata. A binomial GLM with host fixed-effect dummies and cluster-robust standard errors gives OR 1.57 (95% CI [1.02, 2.43]; $p = 0.041$); we prefer the conditional logit because it eliminates the host intercepts rather than estimating many indicators from small strata. The within-host quenched odds ratio is 0.75 (95% CI [0.45, 1.25]; $p = 0.269$; 915 complete cases), and 1.22 without the radius covariate ($p = 0.412$). A host-BGG alignment check supports the satellite interpretation: 56 of 62 non-split CG₄ systems have the same quartet and Lim-host BGG, the 6 misaligned systems are embedded systems whose host BGG is brighter than the compact-group BGG, and all 186 quartet-rank CG₄ satellites are Lim-host satellites; restricting CG₄ to the 56 BGG-aligned systems leaves the three satellite elliptical contrasts stable (against Control_{4C}, OR 1.98).

DS18 classification. The DS18 cross-match of Sect. 2.2 covers 173/248 (70%) of the CG₄ galaxies and 2263/3609 (63%) of the deduplicated controls, i.e. the DR7-era population; it is not worse in CG₄, but neither rate establishes representativeness. With the DS18 $T\text{Type} \leq 0$ outcome and the per-control covariates, standardisation, and physical-group clustering of Sect. 3.1, the CG₄ coefficients are positive in all three contrasts (Table 5); support is broad except that 83% of CG₄ values lie within the RG₄ group-luminosity range,

limiting that smallest comparison. Because DS18 P_{S0} separates S0 from E only within $TType \leq 0$, the environment dependence of the E/S0 mix is left to a dedicated analysis.

Appendix J: Extended size diagnostics

The size-availability counts are included in Fig. H.3; the descriptive mass-size relations and the per-sample sizes predicted at $\log(M_*/M_\odot) = 10.3$ are provided in the online supplement. The concentration index $C = R_{90,r}/R_{50,r}$ provides a model-independent structural proxy: the adjusted satellite coefficient is 0.068 ($p = 0.124$), and the matched concentration difference is -0.008 ($p = 0.904$), offering no evidence for a fixed-mass concentration shift.

The crowding checks mirror Appendix I: excluding galaxies with a projected neighbour within 55 arcsec leaves a satellite coefficient of -0.013 dex ($p = 0.463$), and adding the close-neighbour flag as a covariate gives -0.011 dex ($p = 0.479$).

The per-galaxy difference between the two measures, $d = \log R_{chl,r} - \log R_{50,r}$, averages 0.054 dex, and is larger for elliptical galaxies (0.074 dex) than for spiral galaxies (0.042 dex), as expected if the Petrosian aperture preferentially truncates concentrated profiles. Its rank correlation with the nearest-neighbour separation is $\rho_S = 0.012$ ($p = 0.524$), so residual blending does not drive a strong measure-dependent size systematic.

Adding the pairwise projected tidal index of Sect. 3.4 to the all-galaxy size model changes the CG_4 coefficient from -0.012 dex to 0.005 dex on the pairwise complete-case sample; the local projected-density proxy attenuates the size coefficient in the same way it attenuates the morphology coefficient.

The radial diagnostic is exploratory: mass- and redshift-adjusted satellite size residuals correlate with projected BGG-centric distance with $\rho_S = 0.17$ ($p = 0.026$) in CG_4 and $\rho_S = 0.18$ ($p < 10^{-4}$) in the pooled controls. A more negative inner-region residual would be the tidal-truncation expectation; the observed trends are similar in CG_4 and the controls, so the data do not isolate a CG_4 -specific inner-region size suppression.

Within morphology classes, the satellite-only fits give 0.036 dex for elliptical satellites ($p = 0.068$) and -0.005 dex for spiral satellites ($p = 0.831$), and the $CG_4 \times$ morphology interaction is 0.032 dex ($p = 0.348$). These strata are secondary diagnostics of whether size carries information beyond the morphology shift.

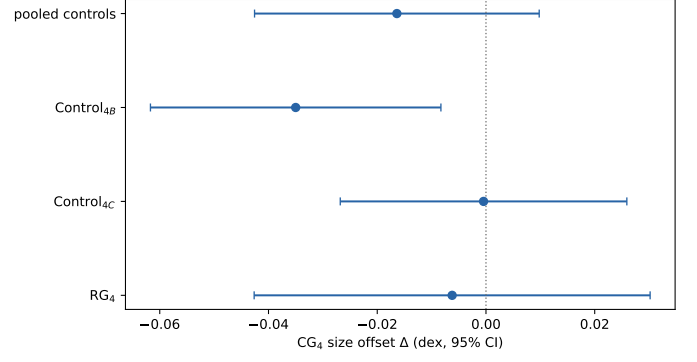


Fig. J.1. Mass- and redshift-adjusted CG_4 size coefficients against the pooled controls and against each control catalogue separately. Bars show 95% confidence intervals; negative values mean smaller CG_4 sizes at fixed mass.